

An Explainable and Uncertainty-Aware Audio-Visual Deepfake Detection Framework using Cross-Modal Attention Fusion

A project report submitted in partial fulfilment of the requirements for the award of the degree of

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in
Computer Science & Engineering (AI & ML)

Under the guidance of
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Submitted by
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CERTIFICATE

*This is to certify that the project entitled "**An Explainable and Uncertainty-Aware Audio-Visual Deepfake Detection Framework using Cross-Modal Attention Fusion**" is a bonafide record of the project presented by **Arya Gopal** (University Register Number: _____), under our supervision and guidance. The project report has been submitted to the Department of Artificial Intelligence and Data Science of RSET, Kakkanad, in partial fulfilment of the requirements for the award of the Degree of Master of Technology in Computer Science & Engineering (AI & ML).*

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I further declare that this work has not been submitted, in part or in full, for the award of any other degree or diploma of this or any other institution, and that all sources of information used in this report have been duly acknowledged.

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ABSTRACT

Deepfake generation techniques can now synthesise highly convincing manipulated audio and video, posing a growing threat to digital trust, forensic investigation and content moderation. Most existing audio-visual deepfake detectors optimise purely for classification accuracy through cross-modal fusion, yet operate as black boxes: they give no human-interpretable explanation for a verdict and no signal of their own confidence, so an incorrect high-confidence decision cannot be distinguished from a reliable one. This work proposes an Explainable and Uncertainty-Aware Audio-Visual Deepfake Detection Framework built around cross-modal attention fusion. Face frames are encoded with EfficientNet-B0 or ViT-B/16 and speech with WavLM-Base+ or XLS-R, and a bidirectional multi-head cross-attention module lets each modality attend to inconsistencies in the other before a classifier produces a real/fake verdict. The framework adds two capabilities largely absent from prior work: joint explainability, using Grad-CAM to highlight manipulated facial regions and SHAP to attribute influential frequency bands and time segments of the audio; and calibrated uncertainty estimation, using Monte Carlo Dropout and Evidential Deep Learning to route low-confidence predictions to a human forensic reviewer. A survey of thirty papers (2016–2026) confirms that fusion accuracy, explainability and uncertainty quantification have advanced independently, but that no existing framework unifies all three for audio-visual deepfakes. The framework is trained on FakeAVCeleb with subject-disjoint splits and evaluated for accuracy, AUC, calibration error, explanation fidelity and cross-dataset generalisation on AV-Deepfake1M and DFD, together with an ablation study and a prototype analyst dashboard. The result is a single pipeline that returns not only a real/fake decision but also the visual and audio evidence for that decision and an honest signal of when it should not be trusted.

Keywords: Audio-visual deepfake detection, cross-modal attention fusion, explainable AI, Grad-CAM, SHAP, uncertainty quantification, Monte Carlo Dropout, Evidential Deep Learning, FakeAVCeleb.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
A-V	Audio-Visual
AUC	Area Under the Curve
CNN	Convolutional Neural Network
DFD	Deepfake Detection Dataset (Google/Jigsaw)
DFD (diagram)	Data Flow Diagram
ECE	Expected Calibration Error
EDL	Evidential Deep Learning
EER	Equal Error Rate
ER	Entity Relationship
FFN	Feed-Forward Network
FV-FA / FV-RA	Fake Video–Fake Audio / Fake Video–Real Audio
GAN	Generative Adversarial Network
GELU	Gaussian Error Linear Unit
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
HLD / LLD	High-Level Design / Low-Level Design
KL	Kullback–Leibler divergence
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Short-Term Memory
MC Dropout	Monte Carlo Dropout
MHA	Multi-Head Attention
MLP	Multi-Layer Perceptron
MTCNN	Multi-task Cascaded Convolutional Networks
ROC	Receiver Operating Characteristic
RV-RA / RV-FA	Real Video–Real Audio / Real Video–Fake Audio
SHAP	SHapley Additive exPlanations
SLS	Sensitive Layer Selection
SRS	System Requirement Specification
SSL	Self-Supervised Learning
TTS	Text-to-Speech
UML	Unified Modeling Language
VAD	Voice Activity Detection
ViT	Vision Transformer
WavLM	Wave Language Model (self-supervised speech model)
XAI	Explainable Artificial Intelligence
XLS-R	Cross-lingual Speech Representations (wav2vec 2.0 based)

Chapter 1

INTRODUCTION

1.1 Background

The last decade has seen remarkable progress in generative artificial intelligence. Generative Adversarial Networks (GANs), diffusion models, neural text-to-speech systems and voice-conversion engines can now synthesise faces, expressions, lip movements and speech that are almost indistinguishable from authentic recordings. The term "deepfake" is used for media in which a person's appearance or voice has been manipulated or synthesised using such deep learning techniques. Early deepfakes were largely visual: the face of one person was swapped onto the body of another in a video, often leaving visible blending artefacts, flickering or inconsistent skin tones. Current generation techniques have removed many of these obvious traces.

At the same time, the barrier to producing such content has fallen sharply. Open-source face-swapping tools, lip-synchronisation models and few-shot voice-cloning services allow a non-expert to create a convincing manipulated video in a short time using commodity hardware. Modern deepfakes are therefore increasingly multimodal: a manipulated or re-enacted face is combined with cloned or synthesised speech so that both what the viewer sees and what the viewer hears support the deception. A detector that inspects only the video frames, or only the audio track, can be evaded by manipulating the other modality alone.

Deepfake detection has consequently become an important research problem in digital media forensics. Early detectors treated the problem as binary image classification, using Convolutional Neural Networks (CNNs) to spot frame-level artefacts. Later work added temporal models to capture flicker and unnatural motion, and separate lines of work addressed synthetic speech detection using spectral features and self-supervised speech models. More recently, audio-visual detectors have been proposed that process both streams and fuse them, exploiting the observation that manipulation frequently breaks the natural correspondence between lip movement and speech.

1.2 Deepfakes and their Societal Impact

The risks posed by deepfakes extend well beyond entertainment. Manipulated videos of political leaders can be circulated during elections to spread disinformation. Fabricated clips can be used to harass or defame individuals. Cloned voices have been used in financial fraud, where an attacker impersonates an executive or relative over a call to authorise a payment. In legal proceedings, the authenticity of video and audio evidence can be challenged, and conversely genuine evidence can be dismissed as fake, a phenomenon often called the "liar's

dividend". Newsrooms and fact-checking organisations must verify user-generated content quickly and defensibly before publication.

Regulators have begun to respond. The European Union's AI Act introduces transparency obligations for synthetic content, and India's IT Rules have been amended to address synthetically generated information on intermediary platforms. These developments increase the demand for detection tools that are not only accurate, but whose decisions can be inspected, justified and audited. A verdict that simply says "fake, 97%" without any supporting evidence is of limited use to a journalist deciding whether to publish, to a moderator deciding whether to remove content, or to a court weighing evidence.

1.3 Why Audio-Visual Fusion

Manipulation artefacts are frequently present in only one modality at a time. In a face-swap video with the original soundtrack, the audio is genuine and all the evidence lies in the visual stream. In a video with a cloned voice dubbed over genuine footage, the frames are authentic and the evidence lies in the audio. In lip-sync attacks, both streams may individually look and sound plausible, yet the mouth movements do not match the spoken phonemes. Figure 1.1 illustrates how comparing the two streams exposes inconsistencies that are invisible when each stream is examined in isolation.

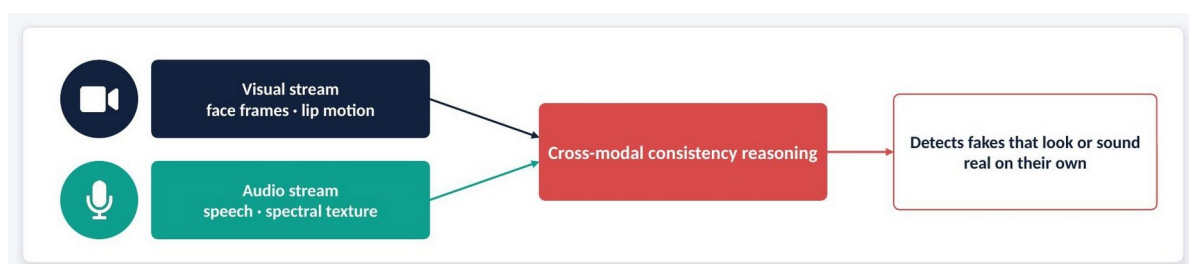


Figure 1.1: Cross-modal consistency reasoning in audio-visual deepfake detection

Cross-modal fusion therefore provides an additional forensic signal: the agreement or disagreement between what is seen and what is heard. Attention-based fusion architectures allow the representation of one modality to query the representation of the other, so that the model can learn which video frames correspond to which segments of speech and where that correspondence is broken.

1.4 Motivation

A review of recent audio-visual deepfake detectors shows that the field has concentrated on raw classification accuracy. Several fusion models report accuracy above 98% on the FakeAVCeleb benchmark. However, three practical weaknesses remain. First, these models behave as black boxes: they provide no indication of which facial region or which part of the audio led to the verdict. Second, they provide no reliable signal of their own confidence; a

wrong prediction is delivered with the same apparent certainty as a correct one. Third, benchmark evaluations have shown that state-of-the-art detectors lose 45–50% of their AUC when tested on in-the-wild deepfakes, which means that confident but wrong predictions are likely in real deployments.

These observations motivate a detector that is simultaneously accurate, explainable across both modalities, and aware of its own uncertainty. Such a system should present the evidence behind every decision and, when it is unsure, should defer to a human forensic analyst rather than issue a silent and possibly wrong verdict.

1.5 Problem Statement

Current audio-visual deepfake detectors rely on opaque late- or attention-based fusion of audio and video streams. They classify a clip as real or fake with high benchmark accuracy but provide no transparent reasoning for the decision and no mechanism for identifying and routing uncertain predictions for human review. In high-stakes domains such as digital forensics, journalism, content moderation and legal evidence authentication, identifying manipulated media reliably and explainably is critical for trust and admissibility.

The problem addressed in this dissertation is therefore: **to design and implement an audio-visual deepfake detection framework that fuses cross-modal evidence using attention, explains each decision with visual and audio saliency, and quantifies predictive uncertainty so that low-confidence predictions are flagged for human review.**

1.6 Objectives

The primary objective of this work is to develop an Explainable and Uncertainty-Aware Audio-Visual Deepfake Detection Framework using Cross-Modal Attention Fusion. The secondary objectives are:

- To study the challenges of existing unimodal and multimodal deepfake detection approaches through a structured literature survey.
- To implement and evaluate unimodal baseline classifiers (a video branch and an audio branch) as reference points.
- To design a dual-branch encoder consisting of a vision backbone for face frames and a self-supervised speech model for audio.
- To design a cross-modal multi-head attention fusion module that combines the two modality-specific representations.
- To integrate Grad-CAM and SHAP based explainability that visualises manipulated facial regions and influential audio segments.

- To design an uncertainty-estimation module based on Monte Carlo Dropout and Evidential Deep Learning that flags low-confidence predictions for human review.
- To evaluate the complete framework on public benchmark datasets, including cross-dataset testing to assess generalisation.

1.7 Scope of the Project

The scope of the work is binary real/fake classification of short audio-visual clips containing a single visible speaker. The framework accepts a video file, extracts the visual and audio streams, and returns a verdict, a calibrated confidence score, an uncertainty estimate, a Grad-CAM heatmap over the face and a SHAP attribution over audio frequency bands. Clips whose confidence falls below a configurable threshold are routed to a review queue.

Fully live, real-time streaming detection, multi-speaker diarisation and temporal localisation of manipulated segments within long videos are outside the scope of this dissertation and are discussed as future work. The work assumes access to the FakeAVCeleb, AV-Deepfake1M and DFD datasets and a CUDA-enabled GPU environment for training and inference.

1.8 Area of Research

This dissertation lies at the intersection of several research areas, each of which contributes a component of the final framework.

1.8.1 Digital Media Forensics

Digital media forensics studies techniques to verify the authenticity and integrity of images, video and audio. Passive forensic methods analyse the content itself, without relying on watermarks or metadata, to identify traces left by manipulation. Deepfake detection is a major contemporary sub-area of passive media forensics, and this work contributes a multimodal, explainable passive detector.

1.8.2 Multimodal Deep Learning

Multimodal learning is concerned with models that process and relate information from different modalities, such as vision, audio and text. Key challenges include representing each modality, aligning them in time and meaning, and fusing them effectively. This work addresses the fusion of visual face features with speech features, a setting in which the two modalities have different sampling rates and statistical properties.

1.8.3 Self-Supervised Speech Representation Learning

Self-supervised speech models such as WavLM and XLS-R are pre-trained on large amounts of unlabelled speech and produce rich contextual representations. These representations have proved highly effective for downstream tasks, including anti-spoofing and synthetic speech

detection, because they capture subtle phonetic and acoustic properties that hand-crafted features may miss.

1.8.4 Attention Mechanisms

Attention mechanisms, popularised by the Transformer architecture, allow a model to weight the importance of different elements of an input when computing a representation. Cross-attention, in which queries come from one sequence and keys and values from another, is a natural tool for relating two modalities. The fusion module in this work uses bidirectional multi-head cross-attention between visual and audio token sequences.

1.8.5 Explainable Artificial Intelligence

Explainable AI (XAI) aims to make the behaviour of machine learning models understandable to humans. Gradient-weighted Class Activation Mapping (Grad-CAM) produces spatial heatmaps showing which image regions influenced a convolutional network's prediction. SHapley Additive exPlanations (SHAP) attributes a prediction to input features using Shapley values from cooperative game theory. This work applies Grad-CAM to the visual branch and SHAP to grouped audio features to produce joint explanations.

1.8.6 Uncertainty Quantification

Uncertainty quantification studies how confident a model should be in its predictions. Standard softmax outputs are often over-confident, particularly on out-of-distribution inputs. Monte Carlo Dropout approximates Bayesian inference by performing multiple stochastic forward passes with dropout enabled, while Evidential Deep Learning models the output as a Dirichlet distribution and derives uncertainty from the total evidence in a single pass. Calibration metrics such as Expected Calibration Error (ECE) measure whether predicted confidence matches observed accuracy.

1.9 Contributions

The main contributions of this dissertation are:

- A structured survey of thirty papers (2016–2026) that maps work on audio-visual fusion, explainability and uncertainty, and identifies the absence of a framework that combines all three for audio-visual deepfakes.
- A dual-branch detector with bidirectional cross-modal multi-head attention fusion of visual (EfficientNet-B0 / ViT-B/16) and audio (WavLM-Base+ / XLS-R) representations.
- A joint explainability module that pairs Grad-CAM heatmaps over face crops with SHAP attributions over grouped Mel-frequency bands and time segments.

- An uncertainty module supporting both Monte Carlo Dropout and an evidential Dirichlet head, together with a threshold-based routing mechanism that sends low-confidence clips to a human review queue.
- A complete system design (SRS, HLD, LLD, DFD, UML and ER diagrams), a prototype analyst dashboard, and an evaluation protocol covering accuracy, calibration, explanation fidelity and cross-dataset generalisation.

1.10 Organisation of the Report

The remainder of this report is organised as follows. Chapter 2 presents the literature survey of thirty papers and identifies the research gap. Chapter 3 describes the proposed system and its design, including the requirement specification, architecture, high-level and low-level design, data flow diagrams, use-case, ER and UML diagrams. Chapter 4 details the implementation: the tools and technologies, the dataset, the preprocessing pipeline, the encoders, the fusion module, the explainability and uncertainty modules, the prototype dashboard and the project execution plan. Chapter 5 presents the evaluation metrics, experimental results, ablation study, calibration analysis, explainability results, cross-dataset evaluation, comparison with existing methods and the test plan. Chapter 6 concludes the report and outlines future work. The appendices contain the presentation slides, the department vision, mission and outcome mapping, the industrial relevance of the work and additional code listings.

Chapter 2

LITERATURE SURVEY

This chapter reviews thirty peer-reviewed papers and preprints published between 2016 and 2026 that are relevant to audio-visual deepfake detection, explainable deepfake detection and uncertainty quantification. For each work, the core method, its main contribution and its limitations with respect to the goals of this dissertation are summarised. The chapter closes with a comparative analysis and a statement of the research gap.

2.1 Survey Methodology

Papers were collected from IEEE Xplore, the ACM Digital Library, SpringerLink, ScienceDirect, the proceedings of NeurIPS, ICML, CVPR and ICASSP, and preprint servers such as arXiv and TechRxiv. The time span 2016–2026 was chosen to include foundational uncertainty work while prioritising 2023–2026 publications for recency. Search keywords included "audio-visual deepfake detection", "cross-modal attention", "multimodal fusion", "explainable AI", "Grad-CAM", "SHAP", "LIME", "Monte Carlo Dropout" and "Evidential Deep Learning".

A paper was included if it satisfied at least one of the following criteria: it proposes an audio-visual fusion architecture for deepfake detection; it proposes or surveys explainability techniques for deepfake detectors; it proposes or surveys uncertainty-quantification methods applicable to deep classifiers; or it introduces a benchmark dataset used in this dissertation. The final set comprises 13 fusion or detection architecture papers, 7 explainability-focused papers, 4 uncertainty-quantification papers and 6 survey or benchmark papers. Figure 2.1 shows this thematic distribution.

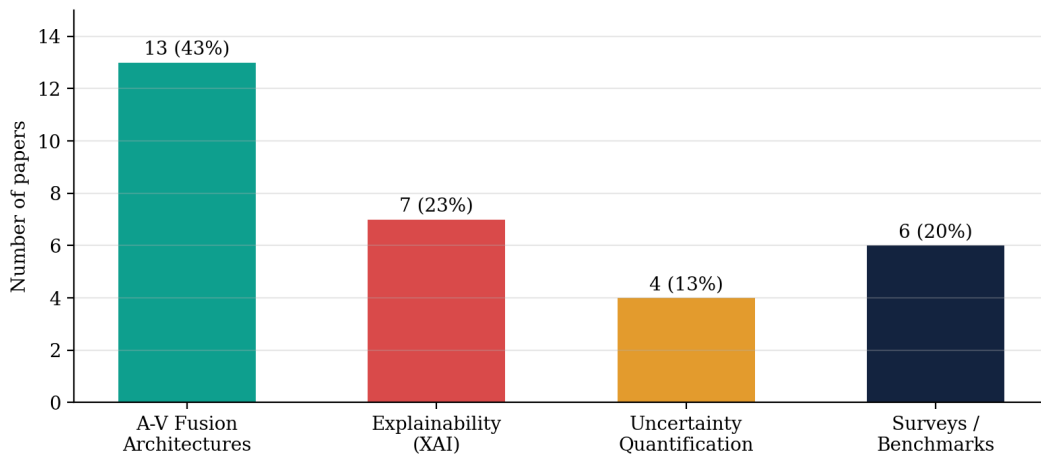


Figure 2.1: Thematic distribution of the thirty surveyed papers

2.2 Audio-Visual Fusion Architectures

This section reviews detectors that process both the visual and the audio streams and combine them to reach a decision.

2.2.1 AVFF: Audio-Visual Feature Fusion for Video Deepfake Detection (Oorloff et al., CVPR 2024)

AVFF proposes a two-stage framework for video deepfake detection that explicitly learns the correspondence between audio and visual streams. In the first stage, the model is trained in a self-supervised manner on real videos using contrastive learning and an autoencoding objective with complementary masking, so that audio features are learned to predict masked visual features and vice versa. In the second stage, the learned audio-visual representations are fine-tuned for real/fake classification.

The approach achieves state-of-the-art performance, reporting 98.6% accuracy on FakeAVCeleb, and demonstrates the value of learning cross-modal correspondence before classification. However, AVFF focuses entirely on detection accuracy. It provides no explanation of which facial region or audio segment triggered a verdict, and its output carries no calibrated confidence estimate, which limits its use in forensic settings where decisions must be justified.

2.2.2 Contextual Cross-Modal Attention for Audio-Visual Deepfake Detection and Localization (Katamneni and Rattani, 2024)

This work introduces a recurrent multi-modal, multi-sequence attention framework (MMMS-BA) that uses contextual cues across audio and visual sequences. The model jointly performs clip-level deepfake detection and temporal localisation of manipulated segments, which is valuable for long videos where only part of the content is manipulated.

The attention weights computed by the model implicitly indicate where cross-modal inconsistencies occur. However, these weights are not translated into human-readable explanations, and the paper does not address calibrated confidence or the routing of uncertain predictions for review.

2.2.3 Cross-Modality and Within-Modality Regularization for Audio-Visual Deepfake Detection (Zou et al., ICASSP 2024)

Zou et al. propose regularisation strategies applied both across and within modalities to improve the robustness of audio-visual fusion. The cross-modality term encourages aligned representations for genuine audio-visual pairs, while the within-modality term preserves modality-specific discriminative information that might otherwise be lost when features are forced into a shared space.

The method improves detection robustness on standard benchmarks. Its contribution is, however, limited to accuracy and robustness; it does not include an explainability layer and does not estimate prediction confidence.

2.2.4 AVFakeNet: A Unified End-to-End Dense Swin Transformer for Audio-Visual Deepfake Detection

AVFakeNet uses a shared dense Swin Transformer backbone to process both audio and visual streams in a unified architecture. The hierarchical, shifted-window attention of the Swin Transformer allows the model to capture local and global patterns in both face frames and audio spectrograms.

While the architecture is powerful, transformer-based decisions remain opaque. The paper does not provide saliency maps or any form of uncertainty quantification, so users cannot inspect why a clip was classified as fake or how reliable the decision is.

2.2.5 MIS-AVoIDD: Modality Invariant and Specific Representation for Audio-Visual Deepfake Detection

MIS-AVoIDD learns two kinds of representation for each modality before fusion: modality-invariant embeddings that capture information shared between audio and video, and modality-specific embeddings that capture information unique to each stream. Combining both types makes the detector more robust when one modality is noisy or manipulated independently.

The paper demonstrates improved robustness, but it does not provide interpretability of the fused decision, nor does it study generalisation across manipulation families in combination with explanation.

2.2.6 DF-TransFusion: Multimodal Deepfake Detection via Lip-Audio Cross-Attention and Facial Self-Attention

DF-TransFusion combines two attention mechanisms: a lip-audio cross-attention module that relates mouth-region features to speech features, and a facial self-attention transformer that models spatial relationships within the face. The cross-attention path is specifically designed to capture lip-sync inconsistencies introduced by lip-sync generation methods.

The approach is closely related to the fusion strategy of this dissertation. However, it lacks a confidence-calibrated verification step for ambiguous cases, and its attention is not converted into explanations presented to an analyst.

2.2.7 Modality-Agnostic Audio-Visual Deepfake Detection

This work proposes an architecture that operates flexibly when either the audio or the visual stream is unavailable or has been manipulated independently. Instead of assuming that both modalities are always present and informative, the model is designed to degrade gracefully and to detect manipulations in whichever modality contains them.

Its focus on robustness to missing or independently manipulated modalities is valuable. The work does not, however, incorporate explainability or uncertainty-based routing of predictions.

2.2.8 A Robust Approach to Multimodal Deepfake Detection

This paper evaluates fusion-based multimodal detectors under realistic perturbations such as video compression, additive audio noise and cross-dataset shifts. It highlights that detectors which perform well on clean benchmark data can degrade significantly under such conditions and proposes design choices that improve robustness.

The study underlines the importance of robustness evaluation, which this dissertation adopts in its cross-dataset experiments. Interpretability and uncertainty calibration are not studied.

2.2.9 MCL: Multimodal Contrastive Learning for Deepfake Detection

MCL applies contrastive learning across audio-visual embeddings. Embeddings of real audio-visual pairs are pulled together in a shared space, while embeddings of fake or mismatched pairs are pushed apart. The resulting embedding distances serve as a signal of audio-visual inconsistency.

The contrastive formulation is effective for learning cross-modal correspondence. However, the learned embeddings are not linked to interpretable regions of the face or segments of the audio, and the model produces no uncertainty output.

2.2.10 CMALDD-PTAF: Cross-Modal Adversarial Learning for Deepfake Detection with Pre-Trained Audio-Visual Features

This work uses pre-trained audio-visual speech representations as input features, combines them in a cross-modality fusion module and applies adversarial training to improve cross-domain performance. Pre-trained features provide strong starting representations, while adversarial training discourages the model from relying on dataset-specific artefacts.

The approach improves generalisation, but at increased computational cost. It does not provide an explanation or confidence-based review mechanism.

2.2.11 KLASSify to Verify: Self-Supervised Audio and Handcrafted Visual Features

KLASSify combines handcrafted visual features, which are interpretable by construction, with self-supervised audio features, and relates them using graph attention. The method reports 92.78% AUC on AV-Deepfake1M++, showing that interpretable visual features can remain competitive when paired with strong audio representations.

The work offers partial interpretability through its handcrafted features, but it does not provide saliency-based explanations for the audio branch or the fused decision and does not include a formal uncertainty module.

2.3 Explainable Deepfake Detection

This section reviews works whose primary contribution is making deepfake detection decisions interpretable.

2.3.1 Explainable Deepfake Detection: Integrating Grad-CAM with CNN-RNN Architecture (Sandeep Kumar et al., 2024)

The authors combine a DenseNet-121 frame encoder with a bidirectional GRU to model temporal dependencies across frames and apply Grad-CAM to highlight the image regions responsible for each prediction. The system operates in real time and reports 88.3% accuracy.

The work demonstrates that Grad-CAM heatmaps can be produced alongside video deepfake predictions at practical speeds. It is, however, unimodal: it analyses video only, provides no audio saliency and does not flag low-confidence predictions.

2.3.2 Deepfake Audio Detection and Justification with Explainable Artificial Intelligence

This paper applies three explanation techniques, LIME, SHAP and Grad-CAM, to audio deepfake detectors that operate on spectrogram representations. By analysing which frequency bands and time regions contribute most to a "fake" decision, the authors provide justifications that relate model behaviour to known artefacts of speech synthesis.

It shows that SHAP-style attribution over frequency bands is meaningful for synthetic speech. The analysis is limited to the audio modality and is not extended to fused audio-visual decisions.

2.3.3 Explainable Deepfake Detection Across Modalities: An Overview

This survey reviews the application of XAI techniques to deepfake detectors for images, video and audio. It categorises methods into saliency-based, perturbation-based and example-based explanations and discusses how each has been used in forensic settings.

The overview is valuable for understanding the landscape, but it is review-only and does not propose a unified cross-modal explainable framework.

2.3.4 Explainable AI for Deepfake Detection: A Grad-CAM Approach to Video Forensics (Salih et al., INCET 2025)

Salih et al. use an EfficientNet-B0 backbone followed by an LSTM to classify videos and apply Grad-CAM to produce heatmaps highlighting manipulated facial regions. The use of EfficientNet-B0 provides a good balance between accuracy and efficiency, and the heatmaps are shown to concentrate on facial areas commonly affected by manipulation.

This work is close to the visual branch of the proposed framework. It is, however, unimodal, performs no audio analysis or fusion and does not provide confidence flagging.

2.3.5 XPlainVerse: A Million-Scale Benchmark for Explainable Deepfake Detection

XPlainVerse introduces a large-scale benchmark in which deepfake samples are paired with grounded natural-language explanations generated with the help of multimodal large language models. The benchmark enables training and evaluation of detectors that output explanations in addition to labels.

The benchmark focuses on image content and does not cover audio-visual fusion or uncertainty-aware verification.

2.4 Uncertainty Quantification

This section reviews works on estimating and calibrating the uncertainty of deep classifiers.

2.4.1 Uncertainty-Aware Deepfake Detection via Multi-View Structural Learning

This work applies Evidential Deep Learning to deepfake detection and combines it with multi-view structural learning. The detector outputs Dirichlet evidence rather than point probabilities, and the reported calibration error on Celeb-DF v2 is reduced by 28% compared with a conventional baseline.

The result shows that evidential methods can substantially improve calibration for deepfake detection. The work is limited to the visual modality and does not include cross-modal fusion or joint explainability.

2.4.2 Prior and Posterior Networks: A Survey on Evidential Deep Learning Methods for Uncertainty Estimation

This survey unifies the family of evidential methods that estimate uncertainty in a single forward pass by predicting the parameters of a prior or posterior distribution over class probabilities. It contrasts these with sampling-based Bayesian approximations and discusses their theoretical properties and practical trade-offs.

The survey provides the theoretical grounding for the evidential head used in this work. It is general-purpose and not applied to deepfake or forensic settings.

2.4.3 Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning (Gal and Ghahramani, ICML 2016)

This foundational paper shows that a neural network trained with dropout can be interpreted as approximate Bayesian inference in a deep Gaussian process. Keeping dropout active at inference time and averaging multiple stochastic forward passes yields a predictive mean and a measure of predictive uncertainty, a technique known as Monte Carlo Dropout.

MC Dropout requires no change to training and is widely used. Its main drawback is the latency of multiple forward passes, and it has not previously been evaluated for multimodal forensic classification.

2.4.4 Evidential Deep Learning to Quantify Classification Uncertainty (Sensoy et al., NeurIPS 2018)

Sensoy et al. propose treating the network outputs as evidence for each class, parameterising a Dirichlet distribution over class probabilities. The total evidence determines the uncertainty, so that a sample with little evidence for any class receives high uncertainty in a single forward pass. A KL-divergence regulariser discourages misleading evidence.

The method provides calibrated uncertainty efficiently, but was demonstrated on generic image classification and not on cross-modal forensic evidence.

2.5 Benchmarks, Surveys and Generalisation Studies

2.5.1 Enhanced Deepfake Detection and Classification Using Vision Transformers

This work trains a custom Vision Transformer for deepfake image detection and reports 92.76% accuracy and an AUC of 0.97. It shows that global self-attention over image patches is effective at capturing manipulation artefacts.

The model operates on single images and therefore uses no temporal, audio or cross-modal information.

2.5.2 Deepfake-Eval-2024: A Multi-Modal In-the-Wild Benchmark of Deepfakes Circulated in 2024

Deepfake-Eval-2024 collects deepfakes that actually circulated on social media and other platforms in 2024 and uses them to evaluate existing detectors. The benchmark shows that state-of-the-art open-source detectors lose roughly 45–50% of their AUC compared with their performance on academic benchmarks.

This finding strongly motivates both cross-dataset evaluation and uncertainty estimation: a detector that will inevitably face out-of-distribution content should be able to indicate when it is unsure. The benchmark itself proposes no detection, explanation or uncertainty method.

2.5.3 Passive Deepfake Detection: A Comprehensive Survey

This survey reviews passive deepfake detection methods across image, video, audio and multimodal settings, organising them by the artefacts they exploit and the architectures they use. It highlights joint audio-visual reasoning as an important direction.

The paper is a survey and does not implement a combined solution.

2.5.4 DeepFake Video Detection: Insights into Model Generalisation

This systematic review compares the generalisation behaviour of CNN-based, transformer-based and multimodal video deepfake detectors when evaluated on datasets or manipulation methods not seen during training. It identifies overfitting to dataset-specific artefacts as a major cause of poor generalisation.

The review is descriptive and does not propose a new architecture, but it informs the evaluation protocol of this dissertation.

2.5.5 A Survey on Multimedia-Enabled Deepfake Detection: State-of-the-Art Tools and Techniques, Emerging Trends

This broad survey covers tools and techniques for multimedia deepfake detection, including audio-visual fusion, and identifies explainability and trust as open research directions.

It supports the research gap identified in this work but does not validate a specific framework.

2.5.6 FakeAVCeleb: A Novel Audio-Video Multimodal Deepfake Dataset (Khalid et al., NeurIPS 2021)

FakeAVCeleb introduces a dataset in which both the video and the audio can be manipulated. Videos of celebrities of diverse ethnic backgrounds are manipulated with face-swapping (FaceSwap, FSGAN), lip-synchronisation (Wav2Lip) and voice cloning (SV2TTS), producing four categories: real video with real audio, fake video with real audio, real video with fake audio, and fake video with fake audio.

FakeAVCeleb has become the standard benchmark for audio-visual deepfake detection and is the primary dataset of this dissertation. Its main limitation is limited manipulation diversity compared with newer diffusion-based generation methods.

2.6 Audio Deepfake and Voice-Clone Detection

2.6.1 Audio Deepfake Detection with Self-Supervised XLS-R and SLS Classifier (Zhang et al., ACM MM 2024)

This paper combines the XLS-R self-supervised speech model with a Sensitive Layer Selection (SLS) classifier that learns to weight the hidden layers of XLS-R most informative for spoof detection. The approach achieves strong performance on audio anti-spoofing benchmarks.

The SLS idea is adopted in the audio branch of this framework. The original work is audio-only and uses no visual cues or fused explanations.

2.6.2 Audio Deepfake Detection with Self-Supervised WavLM and Multi-Fusion Attentive Classifier (ICASSP 2024)

This work uses WavLM representations and an attentive classifier head that fuses information from multiple layers. It shows that WavLM features are highly effective for detecting synthetic speech.

The pipeline is audio-only, with no cross-modal fusion, explainability or uncertainty estimation.

2.6.3 Detection of Cloned Voices in Realistic Forensic Scenarios

This study evaluates automatic speaker recognition and anti-spoofing countermeasures for detecting cloned voices under forensic conditions, such as varying recording channels and durations. It shows that performance can degrade significantly outside controlled laboratory settings.

It focuses on forensic voice comparison rather than fused audio-visual classification.

2.6.4 Voice Cloning and Deepfake Audio Detection Using Deep Learning

This work evaluates MFCC-based detectors and a commercial detection tool against modern voice-cloning attacks and shows that both suffer from generalisation failures on unseen cloning systems.

The paper highlights weaknesses of existing tools but does not propose a cross-modal, explainable or uncertainty-aware solution.

2.7 Comparative Analysis

Table 2.1 summarises all thirty surveyed works by category, key contribution and principal limitation. Table 2.2 compares representative works on five capabilities required by this dissertation: use of audio, use of video, cross-modal fusion, explainability and uncertainty estimation.

Table 2.1: Summary of the thirty surveyed papers

No.	Work	Category	Principal limitation
1	AVFF (CVPR 2024)	A-V fusion	No explanation or confidence
2	Contextual Cross-Modal Attention (MMMS-BA)	A-V fusion	Attention not human-readable
3	Cross/Within-Modality Regularization (ICASSP 2024)	A-V fusion	Accuracy only
4	AVFakeNet (Dense Swin Transformer)	A-V fusion	Opaque; no uncertainty
5	MIS-AVoiDD	A-V fusion	No interpretability
6	DF-TransFusion	A-V fusion	No calibrated verification
7	Modality-Agnostic A-V Detection	A-V fusion	No XAI or routing
8	Robust Multimodal Detection	A-V fusion	No XAI or calibration
9	MCL: Multimodal Contrastive Learning	A-V fusion	Embeddings not interpretable
10	CMALDD-PTAF	A-V fusion	High compute; no XAI
11	KLASSify to Verify	A-V fusion	Partial XAI; no uncertainty
12	Grad-CAM with CNN-RNN	XAI (video)	Unimodal
13	Audio Deepfake Justification with XAI	XAI (audio)	Unimodal
14	XAI Across Modalities: Overview	Survey	Review only
15	Grad-CAM Video Forensics (INCET 2025)	XAI (video)	Unimodal
16	XPlainVerse	XAI benchmark	Images only
17	Multi-View Evidential Learning	Uncertainty	Visual only
18	Prior and Posterior Networks survey	Uncertainty	Not forensic

No.	Work	Category	Principal limitation
19	MC Dropout (ICML 2016)	Uncertainty	Latency; not multimodal
20	Evidential Deep Learning (NeurIPS 2018)	Uncertainty	Generic images
21	Vision Transformer detection	Detection	Images only
22	Deepfake-Eval-2024	Benchmark	No method
23	Passive Deepfake Detection survey	Survey	Review only
24	Insights into Model Generalisation	Survey	No new model
25	Multimedia-Enabled Detection survey	Survey	Review only
26	FakeAVCeleb (NeurIPS 2021)	Dataset	Limited diversity
27	XLS-R + SLS (ACM MM 2024)	Audio detection	Audio-only
28	WavLM + Multi-Fusion Attentive Classifier	Audio detection	Audio-only
29	Cloned Voices in Forensic Scenarios	Audio forensics	Not A-V
30	Voice Cloning & Deepfake Audio Detection	Audio detection	No solution proposed

Table 2.2: Capability comparison of representative works

Work	Audio	Video	Fusion	XAI	Uncertainty
AVFF	✓	✓	✓	✗	✗
Contextual Cross-Modal Attention	✓	✓	✓	✗	✗
Cross/Within-Modality Regularization	✓	✓	✓	✗	✗
AVFakeNet / DF-TransFusion	✓	✓	✓	✗	✗
KLASSify to Verify	✓	✓	✓	Partial	✗
CNN-RNN + Grad-CAM	✗	✓	✗	✓	✗
EfficientNetB0-LSTM + Grad-CAM	✗	✓	✗	✓	✗
Audio XAI (LIME/SHAP/Grad-CAM)	✓	✗	✗	✓	✗
Multi-View Evidential Learning	✗	✓	✗	✗	✓
Proposed framework	✓	✓	✓	✓	✓

2.8 Key Observations

Three observations emerge from the survey:

- **Unimodal detectors** based on CNNs and RNNs detect frame-level visual artefacts with good in-domain accuracy but generalise poorly and ignore audio. Audio-only self-supervised backbones such as XLS-R and WavLM generalise well for speech but ignore visual synchronisation cues. Each unimodal approach is evaluated in isolation from the other modality.
- **Cross-modal fusion models** based on attention (AVFF, MMMS-BA, MIS-AVoiDD) consistently outperform unimodal baselines, with best-case accuracy above 98% on FakeAVCeleb. None, however, reports a calibrated confidence score alongside the verdict, and generalisation to in-the-wild and unseen manipulation types remains weak.

- **Explainability and uncertainty** techniques are well established but have been applied only to unimodal classifiers. Grad-CAM, SHAP and LIME have been used with video-only or audio-only detectors, never jointly, and Monte Carlo Dropout and Evidential Deep Learning have not been paired with audio-visual fusion.

2.9 Research Gap

Based on the survey, the following research gap is identified:

- There is limited integration of cross-modal fusion with dual (audio and visual) explainability.
- No existing work combines attention-based fusion, explainability and uncertainty quantification in a single audio-visual deepfake detection pipeline.
- Explainability research targets single-modality inconsistencies, and generalisation across unseen manipulation types remains weak.
- Cross-dataset and cross-manipulation generalisation is rarely evaluated together with explainability and calibration.

Of the thirty surveyed papers, 43% focus purely on detection accuracy, only 23% address explainability and 13% address uncertainty, in every case for a single modality. None combines all three capabilities with audio-visual fusion. This dissertation addresses this gap by developing a framework that unifies cross-modal attention fusion, joint Grad-CAM and SHAP explainability, and a calibrated MC Dropout / evidential uncertainty module, evaluated across multiple audio-visual deepfake datasets.

Chapter 3

SYSTEM DESIGN

This chapter presents the design of the proposed framework. It begins with an overview of the proposed system, followed by the System Requirement Specification, the overall architecture, the high-level and low-level design, data flow diagrams, the use-case diagram, the ER diagram with its database schema, and the UML class, sequence and activity diagrams. The designs are updated from those submitted at the interim evaluation.

3.1 Proposed System Overview

The proposed system is a dual-branch deep learning framework that analyses the visual and audio streams of a video clip, fuses them with bidirectional cross-modal attention, and produces four outputs for every clip: a real/fake verdict, a calibrated confidence score, an uncertainty estimate and a pair of explanations (a Grad-CAM heatmap over the face and a SHAP attribution over the audio). A decision-routing component compares the confidence and uncertainty with configurable thresholds and either accepts the verdict automatically or sends the clip, together with its evidence, to a human review queue.

The design philosophy follows three principles. First, **fusion by consistency**: rather than concatenating features naively, each modality is allowed to query the other so that mismatches between lip motion and speech become explicit. Second, **evidence with every verdict**: explanations are generated as part of the inference request, so the analyst always sees why a decision was made. Third, **honest confidence**: the system estimates its own uncertainty and prefers deferral to a human over a silent wrong decision.

The processing pipeline consists of the following stages:

- **Data ingestion** reads the input video, validates its format and demultiplexes the visual and audio streams.
- **Preprocessing** detects and aligns faces with MTCNN at 25 frames per second, crops them to 224×224 pixels and samples 32 frames uniformly; the audio is resampled to 16 kHz mono, silence is trimmed with voice-activity detection, and 80-bin log-Mel spectrograms are computed.
- **Encoding** produces a sequence of 512-dimensional visual tokens with EfficientNet-B0 or ViT-B/16 and a sequence of 512-dimensional audio tokens with WavLM-Base+ or XLS-R using Sensitive Layer Selection.
- **Cross-modal fusion** applies multi-head cross-attention in both directions (visual queries over audio, and audio queries over visual), followed by residual connections,

layer normalisation and a feed-forward network, and pools the result into a 1024-dimensional fused vector.

- **Classification** passes the fused vector through a two-layer MLP with softmax to obtain real/fake probabilities.
- **Explanation** computes Grad-CAM on the visual branch and KernelSHAP on grouped Mel bands and time segments.
- **Uncertainty estimation** computes predictive entropy and variance with Monte Carlo Dropout ($T = 30$ passes) or evidential uncertainty from a Dirichlet head.
- **Routing and reporting** apply the confidence threshold, log an audit record and return the verdict with its evidence or place the clip in the review queue.

3.2 System Requirement Specification

3.2.1 Functional Requirements

Table 3.1: Functional requirements

ID	Requirement
FR-01	The system shall accept a video file containing audio and visual streams and extract frame and waveform streams.
FR-02	The system shall validate the input format and reject corrupted files or files without an audio or video stream.
FR-03	The system shall generate a real/fake classification with a confidence score for each clip.
FR-04	The system shall produce a Grad-CAM visual heatmap and a SHAP audio saliency plot for every prediction.
FR-05	The system shall estimate predictive uncertainty using Monte Carlo Dropout or an evidential head.
FR-06	The system shall flag predictions below a configurable confidence threshold, or above an uncertainty threshold, for manual review.
FR-07	The system shall allow a reviewer to approve, reject or escalate a flagged case.
FR-08	The system shall log every inference (input hash, verdict, confidence, uncertainty, model version, timestamp) for audit.
FR-09	The system shall allow an administrator to update the model version and the confidence threshold.
FR-10	The system shall allow an auditor to export audit reports of past decisions.

3.2.2 Non-Functional Requirements

Table 3.2: Non-functional requirements

ID	Category	Requirement
NFR-01	Accuracy	Detection accuracy at least comparable to published state-of-the-art on the FakeAVCeleb test split.
NFR-02	Performance	End-to-end inference latency under 2 seconds per 10-second clip on a single GPU.
NFR-03	Explainability	Explanations generated within the same inference request, without a separate manual step.
NFR-04	Calibration	Expected Calibration Error on the validation set below a defined target (for example 0.05).
NFR-05	Modularity	Fusion, explainability and uncertainty modules independently swappable through configuration.
NFR-06	Reproducibility	Fixed random seeds, versioned configuration files and logged experiments.
NFR-07	Usability	A dashboard that presents verdict, evidence and confidence clearly to a non-expert analyst.
NFR-08	Security & privacy	Uploaded media stored only by hash and path; audit logs protected from modification.

3.2.3 Hardware Requirements

Table 3.3: Hardware requirements

Component	Specification
GPU	NVIDIA CUDA-enabled GPU, 16 GB VRAM or more recommended for training
Processor	Multi-core x86-64 CPU
Memory	16 GB RAM minimum, 32 GB recommended
Storage	SSD with sufficient space for datasets, cached embeddings and checkpoints
Platform	Local workstation, Google Colab or equivalent cloud GPU instance

3.2.4 Software Requirements

Table 3.4: Software requirements

Software	Purpose
Python 3.10	Primary programming language
PyTorch, torchvision, timm	Model development; EfficientNet-B0 and ViT-B/16 backbones
torchaudio, Librosa	Audio loading, resampling and Mel-spectrogram extraction
Hugging Face Transformers	WavLM-Base+ and XLS-R audio encoders
OpenCV, facenet-pytorch (MTCNN)	Frame extraction, face detection and alignment

Software	Purpose
FFmpeg	Demultiplexing audio and video streams
Captum / pytorch-grad-cam, SHAP	Grad-CAM and SHAP explanations
scikit-learn	Metrics, splitting and calibration analysis
Matplotlib, Seaborn	Plots and visualisations
Streamlit, FastAPI	Analyst dashboard and inference API
SQLite	Audit log and review-queue storage
MLflow, Git/GitHub, Docker	Experiment tracking, version control and packaging

3.2.5 Scope, Assumptions and Constraints

- The system performs binary real/fake classification of short audio-visual clips containing a single visible speaker.
- Live streaming detection, multi-speaker diarisation and segment-level localisation are out of scope.
- The FakeAVCeleb, AV-Deepfake1M and DFD datasets are available under their research licences.
- A CUDA-enabled GPU is available for training and inference.
- Input clips contain a detectable frontal or near-frontal face for most frames.

3.3 System Architecture

Figure 3.1 shows the overall system architecture. The input video is split into a visual lane and an audio lane. Each lane is preprocessed and encoded independently. The two token sequences meet in the cross-modal multi-head attention fusion module, whose output feeds the classifier head. The classifier output is passed to the explainability and uncertainty modules, and the combined verdict, confidence and evidence are passed to a decision gate that either auto-accepts the verdict or routes the clip to the human review queue.

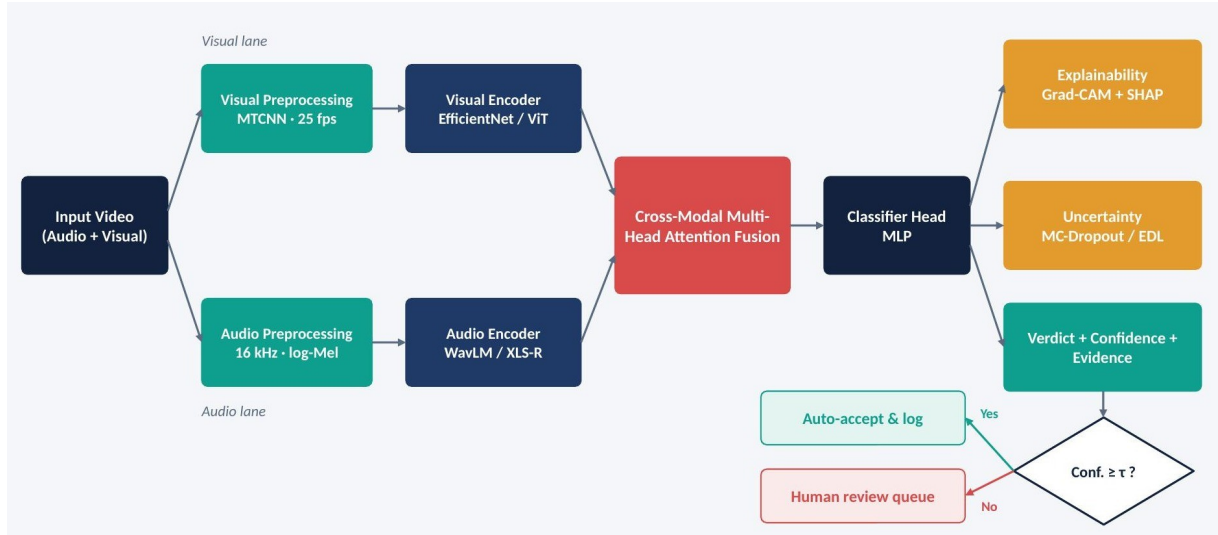


Figure 3.1: Overall system architecture of the proposed framework

3.4 High-Level Design

The high-level design decomposes the system into eight modules with clearly defined responsibilities and interfaces (Figure 3.2).

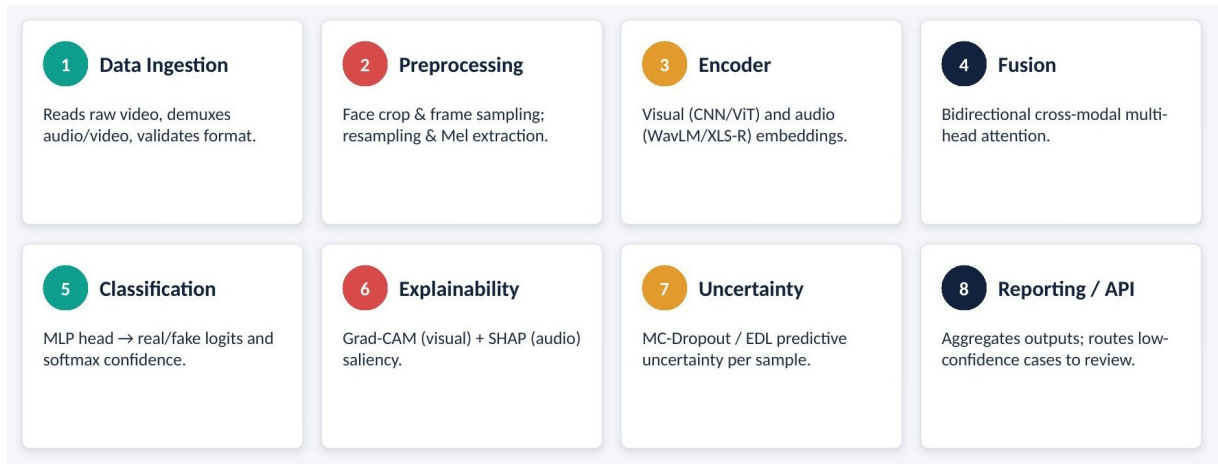


Figure 3.2: High-level design: module breakdown

- **Data Ingestion Module:** reads raw video files, demultiplexes the audio and video streams using FFmpeg and validates the format, duration and presence of both streams.
- **Preprocessing Module:** performs face detection, alignment and cropping, frame sampling and normalisation for the visual stream; resampling, voice-activity detection and log-Mel extraction for the audio stream.
- **Encoder Module:** produces modality-specific token sequences using the visual backbone (EfficientNet-B0 or ViT-B/16) and the audio backbone (WavLM-Base+ or XLS-R with SLS).
- **Fusion Module:** combines the token sequences with bidirectional cross-modal multi-head attention.

- **Classification Module:** maps the fused representation to real/fake logits and softmax confidence.
- **Explainability Module:** generates Grad-CAM heatmaps for the visual stream and SHAP attributions for the audio stream.
- **Uncertainty Module:** estimates predictive uncertainty using MC Dropout or an evidential Dirichlet head.
- **Reporting / API Module:** aggregates the outputs, applies the routing rule, persists audit records and serves results to the dashboard.

3.5 Low-Level Design

Table 3.5 lists the main classes and functions of each module together with the algorithms and parameters they implement.

Table 3.5: Low-level design: module-level classes and algorithms

Module	Class / Function	Algorithm / Detail
Preprocessing	FaceExtractor.crop(), AudioResampler.run()	MTCNN face detection at 25 fps, 224×224 aligned crops, 32 uniformly sampled frames; audio at 16 kHz mono, VAD trimming, 80-bin log-Mel with 25 ms window and 10 ms hop
Visual Encoder	VisualEncoder.forward()	EfficientNet-B0 or ViT-B/16 pretrained on ImageNet; first four blocks frozen; per-frame features projected to 512-d
Audio Encoder	AudioEncoder.forward()	WavLM-Base+ or XLS-R; Sensitive Layer Selection weights hidden layers; projected to 512-d frame tokens
Fusion	CrossAttentionFusion.forward()	Eight-head co-attention: Q = visual, K/V = audio and vice versa; residual + LayerNorm + feed-forward; mean pooling and concatenation to 1024-d
Classifier	ClassifierHead.forward()	Two-layer MLP (1024 → 256 → 2) with GELU and dropout 0.3; softmax output; focal loss
Explainability	GradCAM.generate(), SHAPExplainer.run()	Grad-CAM on the last visual convolution / attention block; KernelSHAP over grouped Mel bands and time segments
Uncertainty	MCDropout.predict(), EDL.dirichlet()	T = 30 stochastic passes giving mean, variance and entropy; or Dirichlet evidence head with uncertainty $u = K / S$
Routing	ReviewQueue.route()	Enqueue if confidence < τ or uncertainty > $u_{\max}$; write audit record to SQLite

3.6 Data Flow Diagrams

3.6.1 DFD Level 0 (Context Diagram)

The context diagram (Figure 3.3) shows the system as a single process interacting with its external entities. The user or forensic analyst uploads a video clip and receives the verdict, explanation and confidence. The model administrator retrains the model and updates the threshold. The dataset store supplies training and evaluation samples, and low-confidence cases and logs are written to the audit log and review queue.

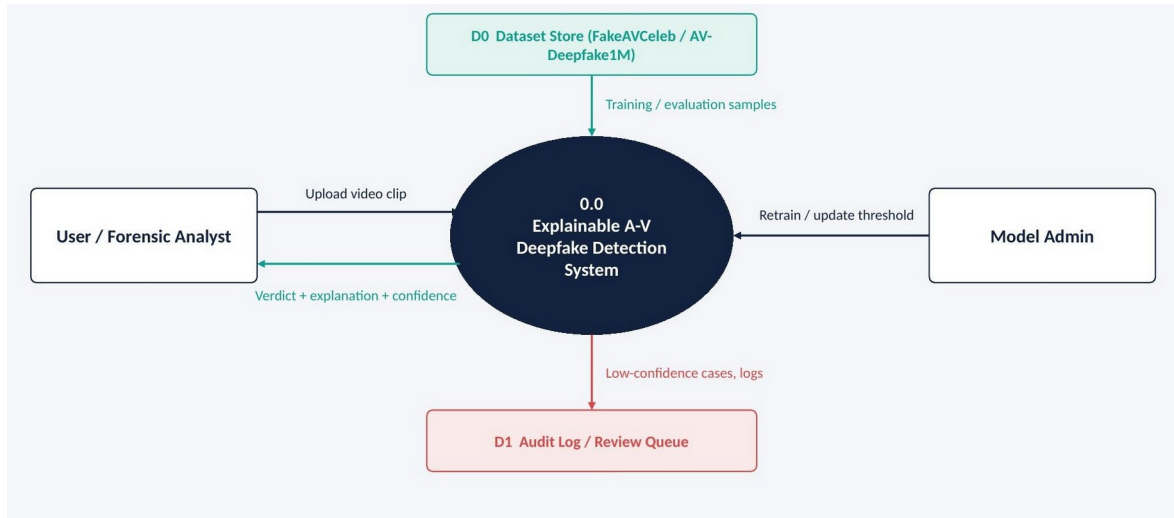


Figure 3.3: DFD Level 0 (context diagram)

3.6.2 DFD Level 1

The Level 1 DFD (Figure 3.4) decomposes the system into six processes: preprocessing of the input (1.0), encoding of audio and visual streams (2.0), cross-modal fusion (3.0), classification (4.0), explanation and uncertainty estimation (5.0) and verdict routing (6.0). Data store D0 holds the datasets, D1 holds the audit log and D2 holds the model weights and the threshold.

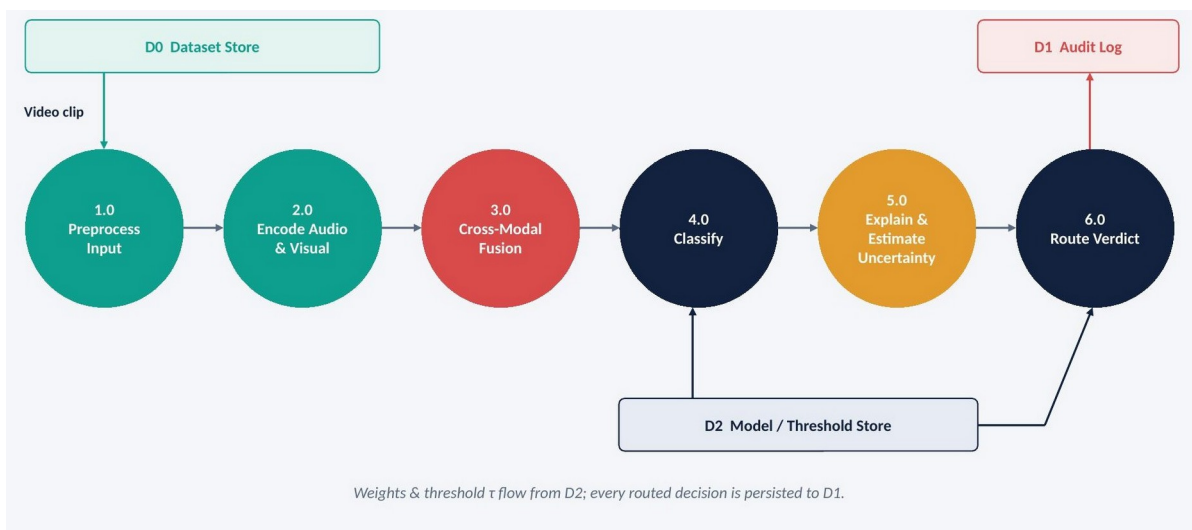


Figure 3.4: DFD Level 1 (process decomposition)

3.7 Use Case Diagram

Figure 3.5 shows the actors and their interactions with the system. The analyst uploads media, views the verdict and confidence, views the explanations and flags cases for manual review. The model administrator retrains or updates the model and sets the confidence threshold. The auditor reviews flagged cases and exports audit reports.

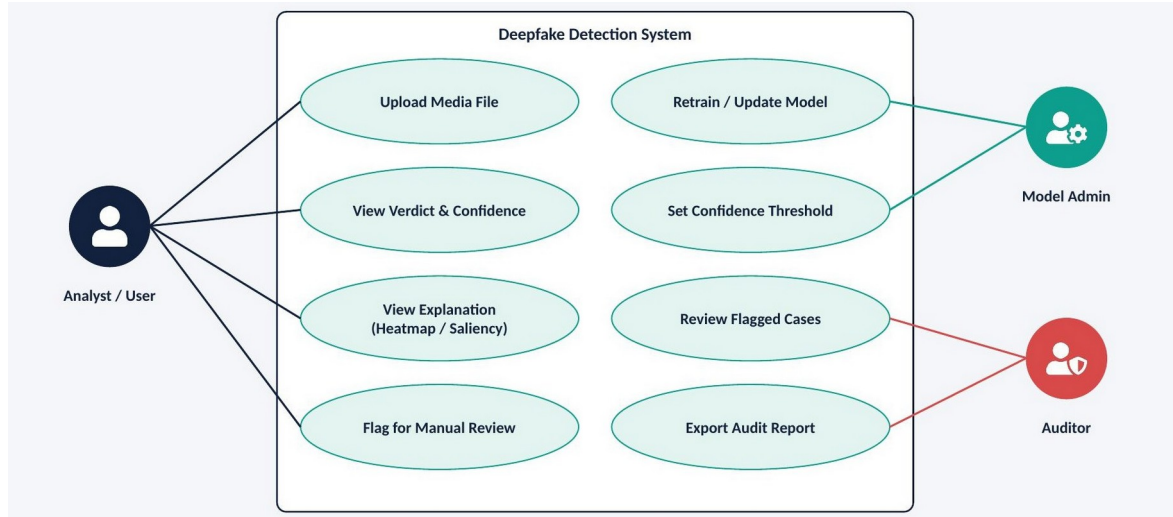


Figure 3.5: Use case diagram

Table 3.6: Use case descriptions

Use case	Actor	Description
Upload media file	Analyst	Uploads a video clip for analysis; system validates format and streams.
View verdict & confidence	Analyst	Views the real/fake verdict, calibrated confidence and uncertainty.
View explanation	Analyst	Views the Grad-CAM heatmap and SHAP audio attribution.
Flag for manual review	Analyst / System	Places a case in the review queue manually or automatically.
Retrain / update model	Admin	Registers a new model version and its metrics.
Set confidence threshold	Admin	Updates τ and $u_{\max}$ used for routing.
Review flagged cases	Auditor	Approves, rejects or escalates flagged cases.
Export audit report	Auditor	Exports decisions and reviewer actions for a date range.

3.8 ER Diagram and Database Design

The ER diagram (Figure 3.6) models the persistent data of the system. A dataset contains many media samples; each media sample may have many inference results produced by a particular model version; each inference result has explanations for one or both modalities and at most one review-queue entry, which is resolved by a user.

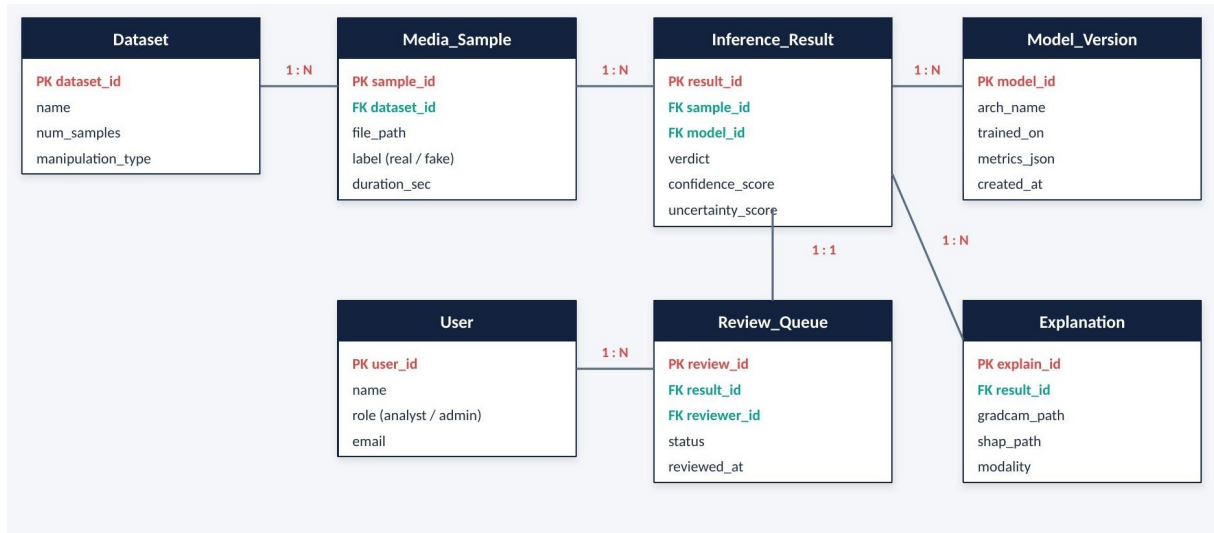


Figure 3.6: ER diagram

Table 3.7: Data dictionary: Media_Sample

Attribute	Type	Description
sample_id (PK)	INTEGER	Unique sample identifier
dataset_id (FK)	INTEGER	Source dataset
file_path	TEXT	Path to stored media
label	TEXT	Ground truth: real / fake (null for new uploads)
duration_sec	REAL	Clip duration in seconds

Table 3.8: Data dictionary: Inference_Result

Attribute	Type	Description
result_id (PK)	INTEGER	Unique result identifier
sample_id (FK)	INTEGER	Analysed media sample
model_id (FK)	INTEGER	Model version used
verdict	TEXT	real / fake
confidence_score	REAL	Calibrated probability of the verdict
uncertainty_score	REAL	Entropy or evidential uncertainty
input_hash	TEXT	SHA-256 hash of the input file
created_at	DATETIME	Timestamp of inference

Table 3.9: Data dictionary: Explanation and Review_Queue

Attribute	Type	Description
explain_id (PK)	INTEGER	Explanation identifier
result_id (FK)	INTEGER	Associated inference result
gradcam_path / shap_path	TEXT	Paths to stored explanation images
modality	TEXT	visual / audio

Attribute	Type	Description
review_id (PK)	INTEGER	Review-queue entry identifier
status	TEXT	pending / approved / rejected / escalated
reviewer_id (FK)	INTEGER	User who resolved the case
reviewed_at	DATETIME	Resolution timestamp

3.9 UML Diagrams

3.9.1 Class Diagram

The class diagram (Figure 3.7) shows the principal classes. MediaSample loads media and extracts streams; Preprocessor crops faces and computes Mel spectrograms; FusionEncoder holds the two encoders and performs cross-attention; DeepfakeClassifier predicts and exposes confidence; ExplainabilityEngine, UncertaintyEstimator and ReviewQueueManager are used by the classifier to explain, quantify uncertainty and route cases respectively.

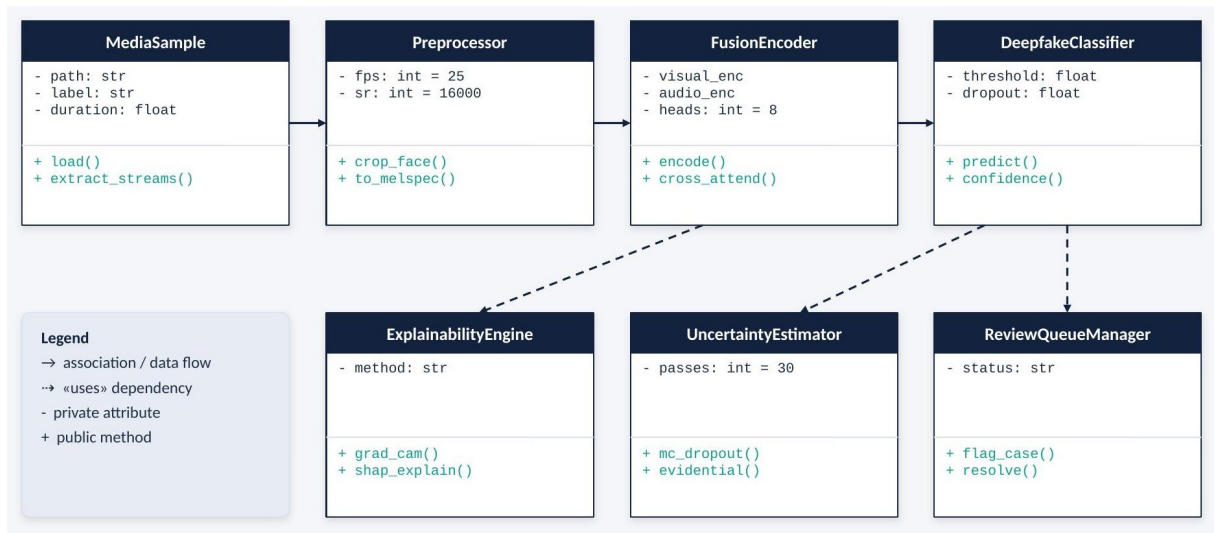


Figure 3.7: UML class diagram

3.9.2 Sequence Diagram

The sequence diagram (Figure 3.8) shows the message flow for one inference request. The user uploads a video to the API, which calls the preprocessor. Once frames and log-Mel features are ready, the API calls the fusion model, which invokes the explainability and uncertainty components. Saliency maps, confidence and uncertainty are returned to the API, which enqueues the case if the confidence is below the threshold and finally returns the verdict, confidence and explanation to the user.

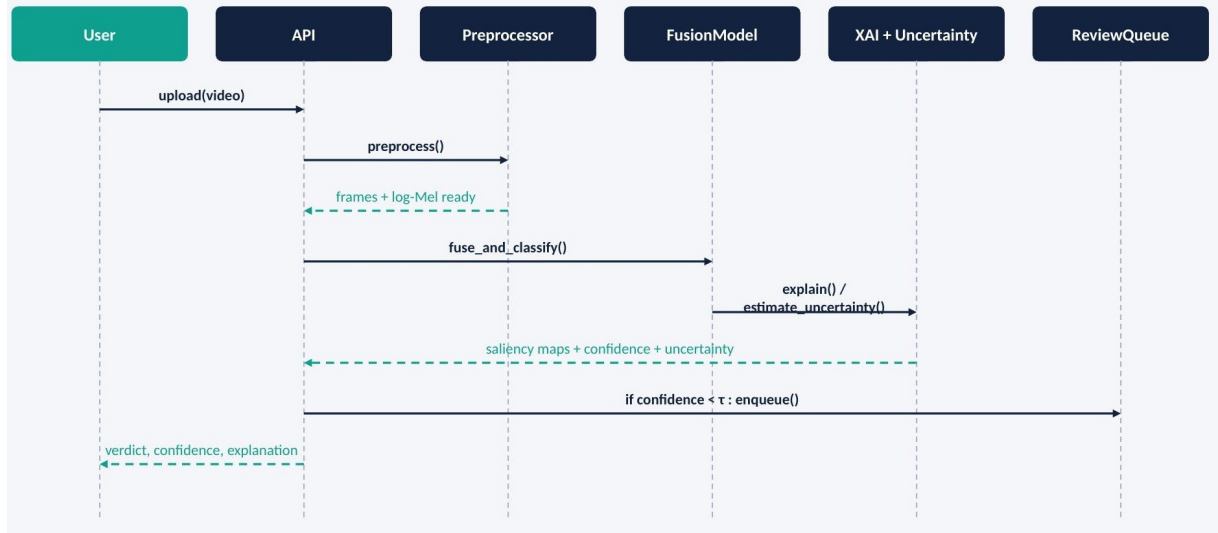


Figure 3.8: UML sequence diagram for the inference flow

3.9.3 Activity Diagram

The activity diagram (Figure 3.9) shows the end-to-end pipeline, including the decision node on confidence. Both the auto-accept path and the human-review path terminate after an audit record is written.

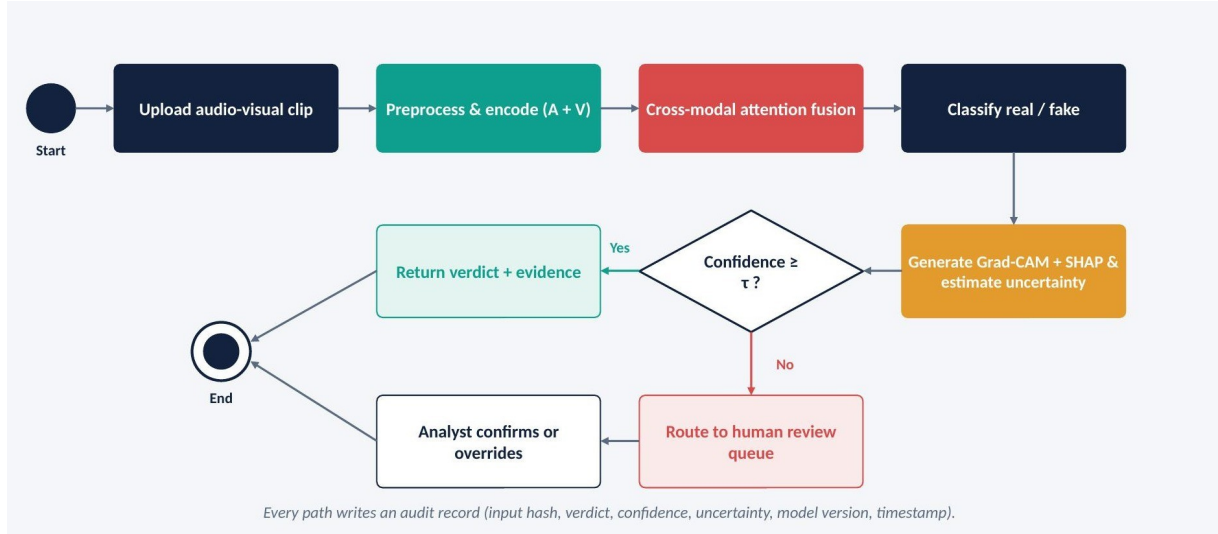


Figure 3.9: UML activity diagram for the end-to-end pipeline

3.10 Detailed Design of the Fusion Module

The fusion module is the core of the framework (Figure 3.10). Let $V \in \mathbb{R}^{(T_v \times d)}$ be the visual token sequence ($T_v = 32$ frames) and $A \in \mathbb{R}^{(T_a \times d)}$ the audio token sequence, with $d = 512$. Scaled dot-product attention is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q K^T}{\sqrt{d_k}}\right) V \quad (3.1)$$

Multi-head attention with $h = 8$ heads projects the inputs into h subspaces of dimension $d_k = d / h = 64$, applies attention in each and concatenates the results. Bidirectional cross-attention is then computed as:

$$Z_v = \text{LayerNorm}(V + \text{MHA}(Q = V, K = A, V = A)) \quad (3.2)$$

$$Z_a = \text{LayerNorm}(A + \text{MHA}(Q = A, K = V, V = V)) \quad (3.3)$$

In Equation 3.2 each visual token attends over the audio sequence, so that each frame gathers the speech evidence most relevant to it; in Equation 3.3 each audio token attends over the frames. The two outputs are mean-pooled over time and concatenated to form the fused representation:

$$h = [\text{pool}(Z_v) ; \text{pool}(Z_a)] \in \mathbb{R}^{1024} \quad (3.4)$$

The fused vector is passed through a feed-forward layer and the classifier head. Because both directions are modelled, a manipulation that affects only the video (face swap with real audio) or only the audio (voice clone over real video) produces an atypical attention pattern in at least one direction.

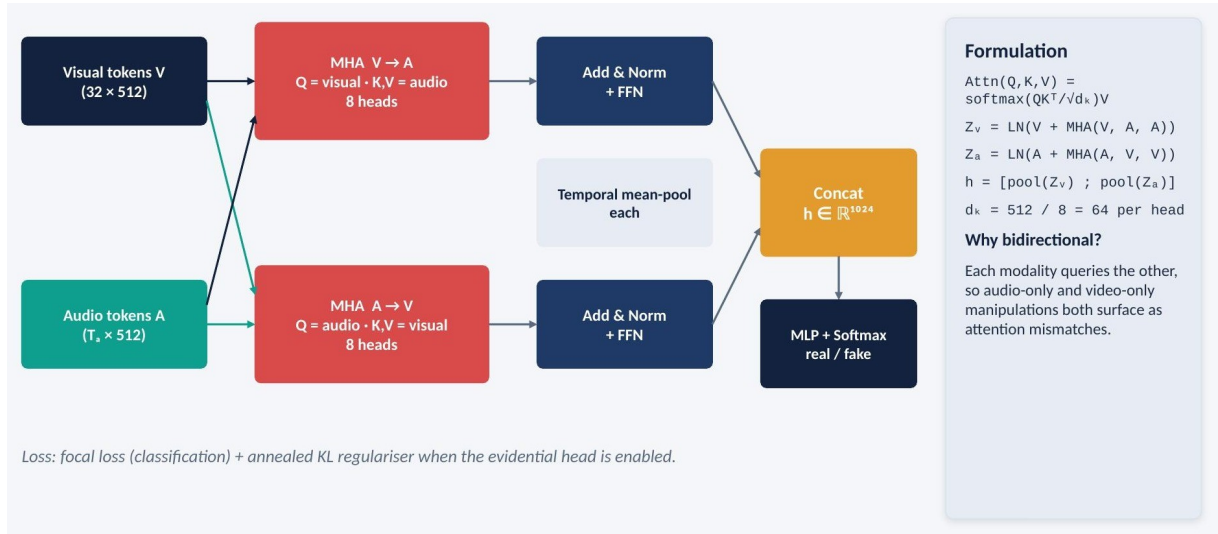


Figure 3.10: Detailed architecture of the cross-modal multi-head attention fusion module

3.11 Design of the Decision-Routing Mechanism

Figure 3.11 shows the design of the human-in-the-loop routing mechanism. For each clip, the system computes the confidence of the predicted class and an uncertainty score. The clip is automatically accepted only if the confidence is at least τ and the uncertainty is at most u_max ; otherwise it is placed in the review queue. Both thresholds are tuned on the validation set to balance reviewer workload against the residual error rate of auto-accepted decisions. Every decision, whether automatic or reviewed, is logged for audit.

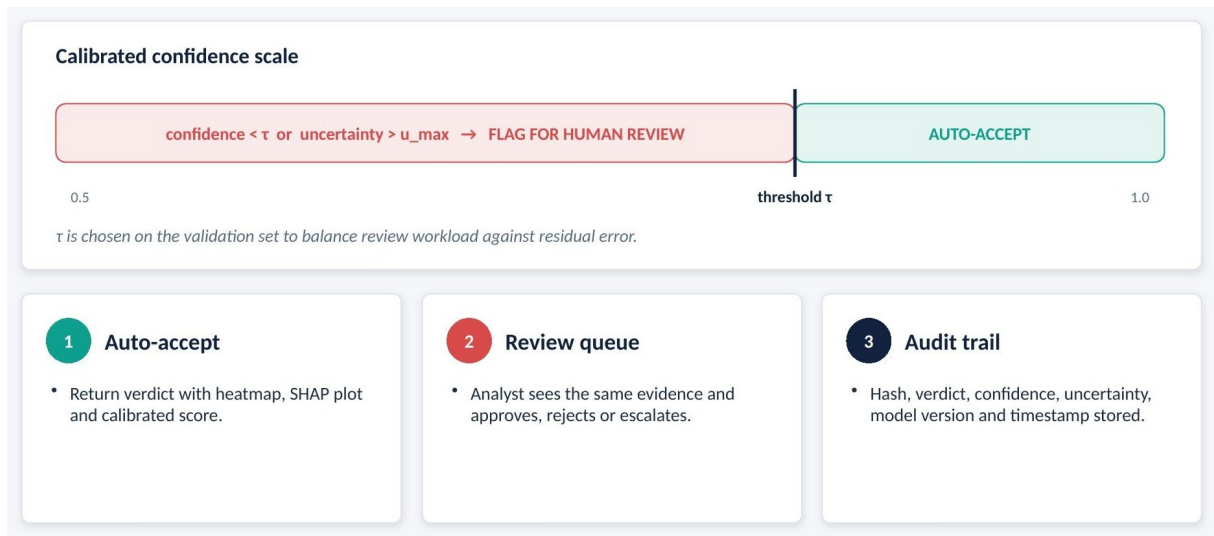


Figure 3.11: Design of the human-in-the-loop decision routing

Chapter 4

IMPLEMENTATION

This chapter describes how the designed framework was implemented. It covers the tools and technologies used, the dataset, and the module-wise implementation of preprocessing, encoding, fusion, classification, explainability, uncertainty estimation, decision routing and the analyst dashboard, with key code snippets. It closes with the training protocol, the project structure and the project execution plan.

4.1 Tools and Technologies

4.1.1 Core Development Environment

- **Python 3.10:** the primary language for data processing, model construction, training, evaluation and the dashboard. Its scientific ecosystem provides mature libraries for every stage of the pipeline.
- **Jupyter Notebook / Google Colab:** used for experimentation, data exploration and visualisation, and for GPU-backed training runs.
- **Visual Studio Code:** used for developing the modular source code, the API and the dashboard.

4.1.2 Deep Learning Frameworks

- **PyTorch:** the core framework. Its dynamic computation graph simplifies the implementation of custom modules such as the bidirectional cross-attention block and makes it straightforward to switch dropout on at inference time for MC Dropout.
- **torchvision and timm:** provide ImageNet-pretrained EfficientNet-B0 and ViT-B/16 backbones for the visual encoder.
- **Hugging Face Transformers:** provides pretrained WavLM-Base+ and XLS-R models and access to all hidden layers, which is required for Sensitive Layer Selection.

4.1.3 Media Processing

- **FFmpeg:** demultiplexes audio and video streams and converts audio to 16 kHz mono WAV.
- **OpenCV:** reads video frames, resizes and converts colour spaces.
- **facenet-pytorch (MTCNN):** detects faces and facial landmarks for alignment and cropping.

- **torchaudio and Librosa:** load and resample audio, compute log-Mel spectrograms and perform voice-activity detection.

4.1.4 Explainability and Uncertainty

- **Captum / pytorch-grad-cam:** compute Grad-CAM heatmaps for the visual branch.
- **SHAP:** computes KernelSHAP attributions over grouped Mel-band and time-segment features.
- **Custom PyTorch code:** implements MC Dropout sampling, the evidential Dirichlet head, the KL regulariser and calibration metrics.

4.1.5 Evaluation, Visualisation and Deployment

- **scikit-learn:** computes accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices, and performs subject-disjoint data splitting.
- **Matplotlib and Seaborn:** produce training curves, ROC curves, reliability diagrams and explanation overlays.
- **MLflow:** logs parameters, metrics and artefacts for each experiment to support reproducibility.
- **Streamlit and FastAPI:** implement the analyst dashboard and the inference API.
- **SQLite:** stores the audit log and the review queue.
- **Git/GitHub and Docker:** provide version control and a reproducible runtime environment.

4.2 Dataset Description

The primary dataset is FakeAVCeleb, an audio-visual multimodal deepfake dataset introduced at the NeurIPS 2021 Datasets and Benchmarks track. It contains approximately 20,000 video clips with paired audio, derived from videos of celebrities from diverse ethnic backgrounds. Visual manipulations are produced with FaceSwap and FSGAN, lip-synchronisation manipulations with Wav2Lip, and audio manipulations with SV2TTS voice cloning. The resulting clips fall into the four categories shown in Figure 4.1.

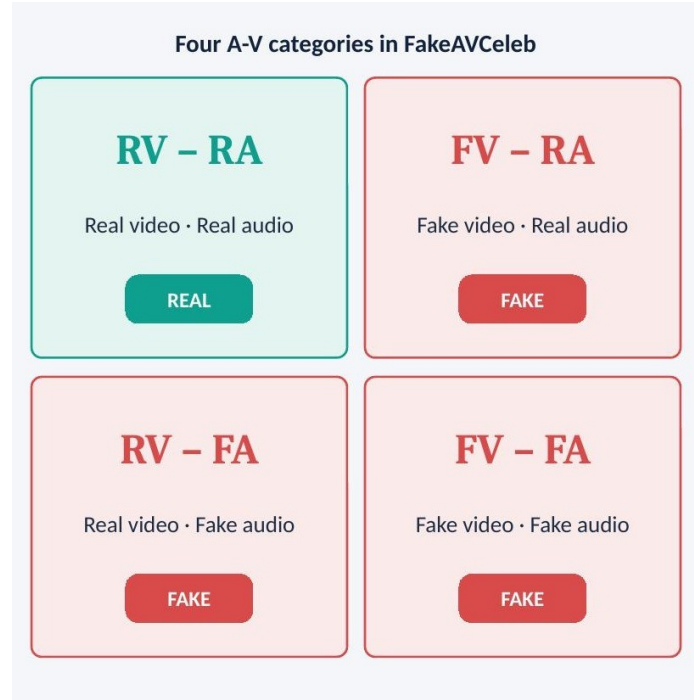


Figure 4.1: Four audio-visual categories in FakeAVCeleb

Table 4.1: Dataset properties

Property	Description
Primary dataset	FakeAVCeleb (audio-visual multimodal deepfake dataset)
Size	Approximately 20,000 clips with paired audio
Manipulations	FaceSwap, FSGAN, Wav2Lip lip-sync, SV2TTS voice cloning
Categories	RV-RA (real), FV-RA, RV-FA, FV-FA (fake)
Task	Binary classification: real vs fake
Split	70% training, 15% validation, 15% test; subject-disjoint
Class balance	Real-video/real-audio clips form a small minority; handled with focal loss and class-balanced sampling
Cross-dataset sets	AV-Deepfake1M (audio-visual) and DFD (visual manipulation) for generalisation testing

A clip is labelled real only if both its video and its audio are genuine; any manipulation in either stream makes the clip fake. The dataset is split in a subject-disjoint manner so that no identity appears in more than one of the training, validation and test sets. This prevents the model from achieving high accuracy by memorising identities and gives a more honest estimate of generalisation. AV-Deepfake1M and DFD are used only for testing, never for training.

4.3 Preprocessing Module

The preprocessing module prepares the visual and audio streams in two parallel lanes (Figure 4.2).

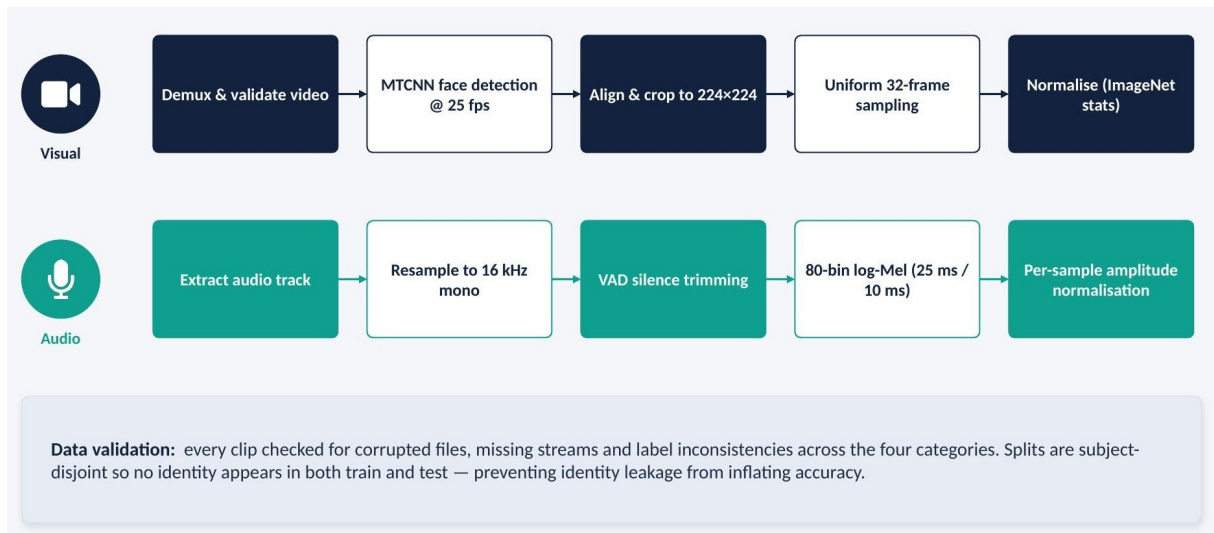


Figure 4.2: Preprocessing pipeline for the visual and audio lanes

4.3.1 Data Validation

Before preprocessing, every clip is checked for file corruption, missing audio or video streams, zero-length streams and label inconsistencies across the four categories. Clips that fail validation are logged and excluded. A manifest file records the path, label, category, subject identifier and duration of every valid clip, and is used to generate the subject-disjoint splits.

4.3.2 Visual Preprocessing

Frames are read at 25 fps. MTCNN detects the face and five facial landmarks in each frame; the face is aligned using the eye positions, cropped with a small margin and resized to 224×224 pixels. To bound computation, 32 frames are sampled uniformly across the clip. Pixel values are normalised with ImageNet mean and standard deviation. Face crops are cached to disk so that repeated experiments do not repeat detection.

4.3.3 Audio Preprocessing

The audio track is extracted with FFmpeg and resampled to 16 kHz mono. A voice-activity-detection mask removes leading, trailing and long internal silences. For the classical baseline and for SHAP analysis, 80-bin log-Mel spectrograms are computed with a 25 ms window and 10 ms hop. The raw normalised waveform is passed to the self-supervised encoder, which computes its own internal features.

Listing 4.1: Preprocessing: face extraction and log-Mel computation

```

mtcnn = MTCNN(image_size=224, margin=20, post_process=False, device=dev)

def extract_faces(video_path, n_frames=32, fps=25):
    frames = read_frames(video_path, fps=fps)          # OpenCV
    idx = np.linspace(0, len(frames) - 1, n_frames).astype(int)
    faces = [mtcnn(frames[i]) for i in idx]
    faces = [f for f in faces if f is not None]
    return torch.stack(faces) / 255.0                  # (T, 3, 224, 224)

```

```
def log_mel(wav, sr=16000):
    mel = torchaudio.transforms.MelSpectrogram(
        sample_rate=sr, n_fft=400, hop_length=160, n_mels=80)(wav)
    return torch.log(mel + 1e-6) # (80, frames)
```

4.4 Dual-Branch Encoders

Figure 4.3 summarises the two encoders that convert the preprocessed streams into token sequences of equal width.

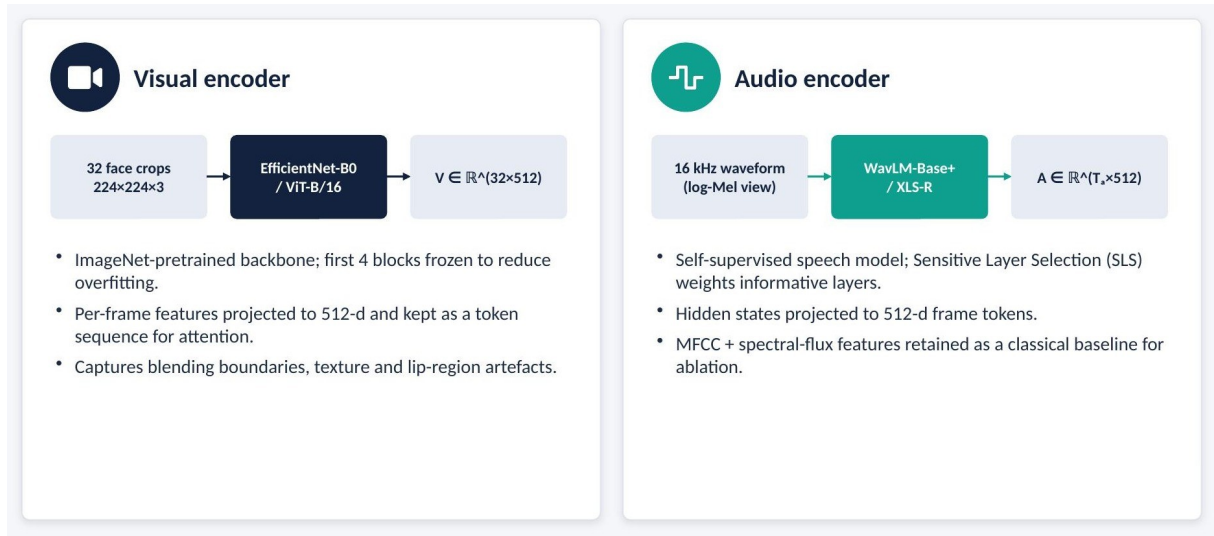


Figure 4.3: Visual and audio encoders

4.4.1 Visual Encoder

The visual encoder uses an ImageNet-pretrained EfficientNet-B0 (or, as an alternative configuration, ViT-B/16). The first four blocks are frozen to reduce overfitting on the limited number of subjects, while the later blocks are fine-tuned. The pooled feature of each frame is projected by a linear layer to 512 dimensions, yielding a token sequence V of shape 32×512 . Keeping one token per frame, rather than pooling over time, allows the fusion module to relate individual frames to the audio.

4.4.2 Audio Encoder

The audio encoder uses WavLM-Base+ (or XLS-R). Different transformer layers of self-supervised speech models capture different information: lower layers are more acoustic, higher layers more phonetic. Following the Sensitive Layer Selection approach, the hidden states of all layers are combined with learned softmax weights so that the model can emphasise the layers most sensitive to synthesis artefacts. The combined representation is projected to 512 dimensions, yielding a token sequence A of shape $T_a \times 512$, where T_a depends on the clip length (approximately 50 tokens per second).

Listing 4.2: Audio encoder with Sensitive Layer Selection

```
class AudioEncoder(nn.Module):
    def __init__(self, name="microsoft/wavlm-base-plus", d=512):
        super().__init__()
        self.ssl = WavLMModel.from_pretrained(name, output_hidden_states=True)
        L = self.ssl.config.num_hidden_layers + 1
        self.layer_w = nn.Parameter(torch.zeros(L)) # SLS weights
        self.proj = nn.Linear(self.ssl.config.hidden_size, d)

    def forward(self, wav):
        hs = torch.stack(self.ssl(wav).hidden_states) # (L,B,Ta,768)
        w = self.layer_w.softmax(0)[: , None, None, None]
        return self.proj((w * hs).sum(0)) # (B,Ta,512)
```

4.5 Cross-Modal Attention Fusion and Classification

The fusion module implements Equations 3.1–3.4 using two `nn.MultiheadAttention` layers with eight heads each. Residual connections and layer normalisation stabilise training. The pooled outputs are concatenated to a 1024-dimensional vector, passed through a feed-forward layer and then through the classification head. The attention weights of both directions are returned so that they can be inspected alongside the Grad-CAM and SHAP explanations.

Listing 4.3: Cross-modal attention fusion module

```
class CrossAttentionFusion(nn.Module):
    def __init__(self, d=512, heads=8, p=0.3):
        super().__init__()
        self.v2a = nn.MultiheadAttention(d, heads, dropout=p, batch_first=True)
        self.a2v = nn.MultiheadAttention(d, heads, dropout=p, batch_first=True)
        self.norm_v, self.norm_a = nn.LayerNorm(d), nn.LayerNorm(d)
        self.ffn = nn.Sequential(nn.Linear(2*d, 2*d), nn.GELU(), nn.Dropout(p))
        self.head = nn.Sequential(nn.Linear(2*d, 256), nn.GELU(),
                                   nn.Dropout(p), nn.Linear(256, 2))

    def forward(self, V, A):
        # V:(B,Tv,512) A:(B,Ta,512)
        zv, w_va = self.v2a(V, A, A) # visual queries attend to audio
        za, w_av = self.a2v(A, V, V) # audio queries attend to video
        zv = self.norm_v(V + zv).mean(1)
        za = self.norm_a(A + za).mean(1)
        h = self.ffn(torch.cat([zv, za], dim=-1)) # fused 1024-d
        return self.head(h), (w_va, w_av)
```

4.5.1 Loss Function

Because real clips form a small minority of FakeAVCeleb, the classifier is trained with focal loss, which down-weights easy, well-classified examples and focuses learning on hard ones:

$$FL(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \quad (4.1)$$

where p_t is the predicted probability of the true class, α_t is a class-balancing weight and γ is the focusing parameter. Class-balanced sampling is additionally used when constructing training batches.

4.6 Explainability Module

Figure 4.4 summarises the explainability module, which produces one explanation per modality.

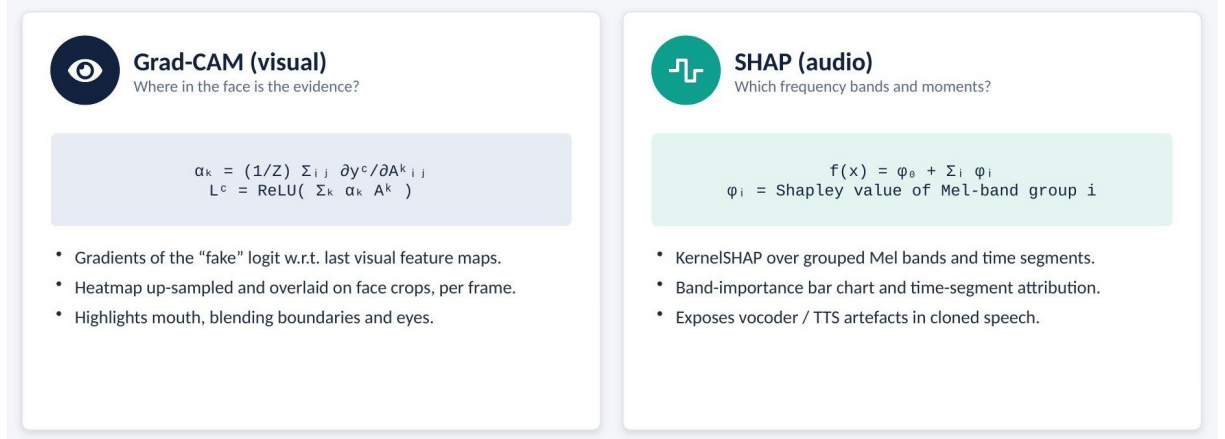


Figure 4.4: Explainability module: Grad-CAM for the visual stream and SHAP for the audio stream

4.6.1 Grad-CAM for the Visual Stream

For the predicted "fake" class c , Grad-CAM computes the gradient of the class score y^c with respect to the feature maps A^k of the last convolutional block of the visual encoder, averages these gradients spatially to obtain the importance weight of each map, and forms a weighted combination:

$$\alpha_k^c = (1/Z) \sum_i \sum_j \partial y^c / \partial A^k_{ij} \quad (4.2)$$

$$L^c = \text{ReLU}(\sum_k \alpha_k^c A^k) \quad (4.3)$$

Because the classifier sits after the fusion module, the gradients flow through the cross-attention layers, so the heatmap reflects the visual evidence as used by the fused decision rather than by the visual branch alone. The heatmap for each frame is up-sampled to 224×224 and overlaid on the face crop, and the frames with the highest activation are shown in the dashboard.

4.6.2 SHAP for the Audio Stream

SHAP explains a prediction $f(x)$ as a sum of feature contributions φ_i added to a base value φ_0 :

$$f(x) = \varphi_0 + \sum_i \varphi_i \quad (4.4)$$

Computing exact Shapley values over thousands of spectrogram bins is infeasible, so the log-Mel spectrogram is divided into groups: eight frequency bands and a fixed number of time segments. KernelSHAP estimates the contribution of each group by masking groups (replacing

them with a background value from real clips) and observing the change in the fused model's output. The result is presented as a band-importance bar chart and a time-segment attribution strip.

Listing 4.4: Grad-CAM and grouped KernelSHAP

```
cam = GradCAM(model=vis_wrapper, target_layers=[visual_enc.backbone.blocks[-1]])
heatmaps = cam(input_tensor=faces, targets=[ClassifierOutputTarget(FAKE)])

def shap_audio(model, V, mel, n_bands=8, n_segs=10, nsamples=300):
    groups = make_groups(mel.shape, n_bands, n_segs)  # masks per group
    def f(z):
        x = apply_masks(mel, groups, z, background=real_mean_mel)
        return model.predict_from_mel(V, x)[: , FAKE].cpu().numpy()
    expl = shap.KernelExplainer(f, np.zeros((1, len(groups))))
    return expl.shap_values(np.ones((1, len(groups))), nsamples=nsamples)
```

4.6.3 Evaluating Explanation Fidelity

Explanations are evaluated quantitatively using deletion and insertion curves. For deletion, the most important pixels (or Mel groups) according to the explanation are progressively removed and the drop in the "fake" probability is recorded; a faithful explanation causes a steep drop and hence a low deletion AUC. For insertion, the most important regions are progressively added to a blurred or silent input, and a faithful explanation yields a high insertion AUC. The stability of SHAP attributions is measured by repeating the estimation with different random seeds.

4.7 Uncertainty Module

Figure 4.5 summarises the two uncertainty methods implemented.

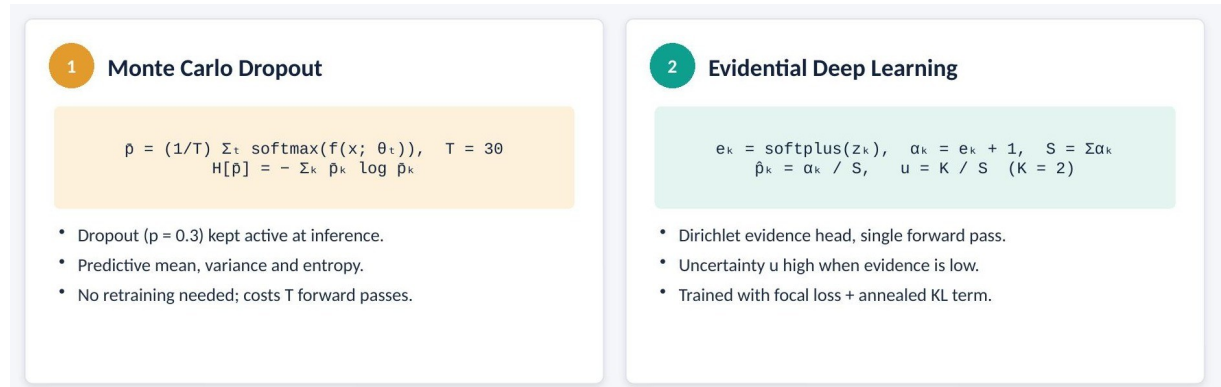


Figure 4.5: Uncertainty module: Monte Carlo Dropout and Evidential Deep Learning

4.7.1 Monte Carlo Dropout

At inference time only the dropout layers are switched to training mode, while batch-normalisation layers remain in evaluation mode. The model is run $T = 30$ times on the same input, and the predictive mean, variance and entropy are computed:

$$\bar{p} = (1/T) \sum_t \text{softmax}(f(x; \theta_t)) \quad (4.5)$$

$$H[\bar{p}] = - \sum_k \bar{p}_k \log \bar{p}_k \quad (4.6)$$

To reduce latency, the encoders are run once and only the fusion module and classifier (which contain the dropout layers) are sampled T times.

4.7.2 Evidential Deep Learning

In the evidential configuration, the classifier outputs non-negative evidence $e_k = \text{softplus}(z_k)$ for each class k . The Dirichlet parameters are $\alpha_k = e_k + 1$, the total strength is $S = \sum_k \alpha_k$, the expected probability is $\hat{p}_k = \alpha_k / S$ and the uncertainty is:

$$u = K / S \quad (4.7)$$

where $K = 2$ is the number of classes. The evidential head is trained with the classification loss plus a KL-divergence term that shrinks misleading evidence towards the uniform Dirichlet, with a coefficient annealed from zero during early epochs.

Listing 4.5: MC Dropout inference, evidential head and routing

```
def enable_dropout(model):
    for m in model.modules():
        if isinstance(m, nn.Dropout):
            m.train()                    # BatchNorm stays in eval

@torch.no_grad()
def mc_predict(fusion, V, A, T=30):
    fusion.eval(); enable_dropout(fusion)
    probs = torch.stack([fusion(V, A)[0].softmax(-1) for _ in range(T)])
    mean = probs.mean(0)
    ent = -(mean * mean.clamp_min(1e-8).log()).sum(-1)
    return mean, probs.var(0).sum(-1), ent

def evidential(logits, K=2):
    alpha = F.softplus(logits) + 1
    S = alpha.sum(-1, keepdim=True)
    return alpha / S, K / S.squeeze(-1)

def route(mean, unc, tau=0.85, u_max=0.5):
    conf, pred = mean.max(-1)
    review = (conf < tau) | (unc > u_max)
    return pred, conf, review           # review=True -> human queue
```

4.7.3 Threshold Calibration

The routing thresholds τ and $u_{\max}$ are selected on the validation set. For a grid of candidate values, the fraction of clips routed to review (coverage cost) and the error rate among auto-accepted clips (selective risk) are computed. The operating point is chosen to keep the selective error below a target while limiting the review workload. Expected Calibration Error is computed before and after uncertainty modelling to verify that confidence scores are meaningful.

4.8 Decision Routing, Audit Logging and API

The FastAPI service exposes an endpoint that accepts a video upload, runs the full pipeline and returns a JSON response containing the verdict, confidence, uncertainty, routing decision and paths to the explanation images. Every request writes an `Inference_Result` record and associated `Explanation` records to SQLite; flagged cases also create a `Review_Queue` entry with status "pending". Reviewers update the status through the dashboard, and each change is timestamped.

4.9 Analyst Dashboard Prototype

The Streamlit dashboard (Figure 4.6) allows an analyst to upload a clip and view the results in a single screen: the video preview with the Grad-CAM overlay and waveform, the verdict with its confidence, uncertainty and model version, the SHAP band-importance chart, a confidence gauge showing the threshold, and the review queue with actions to approve, reject or escalate cases.

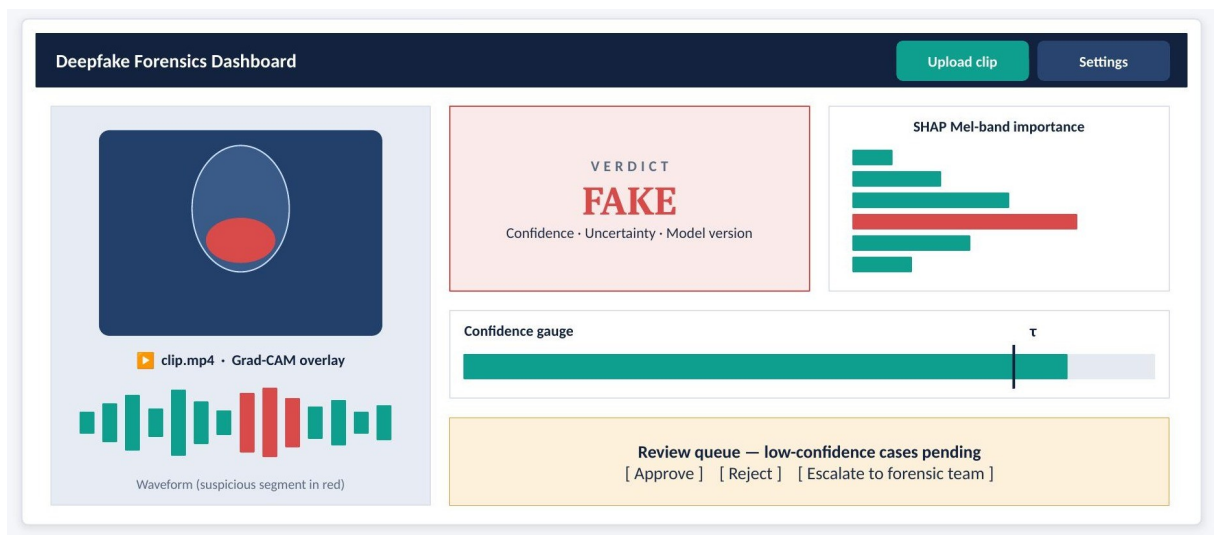


Figure 4.6: Layout of the analyst dashboard prototype

4.10 Training Protocol

Table 4.2 lists the training configuration. Models are trained with AdamW and a cosine learning-rate schedule. The checkpoint with the best validation F1-score is retained. All runs use fixed random seeds and are logged in MLflow with their configuration files.

Table 4.2: Training configuration and hyper-parameters

Hyper-parameter	Value
Optimiser	AdamW
Learning rate	2×10^{-4} with cosine decay
Batch size	32
Attention heads	8

Hyper-parameter	Value
Embedding size	512 per modality, 1024 fused
Dropout	0.3 in fusion and classifier
MC Dropout passes	T = 30 at inference
Loss	Focal loss (+ annealed KL term for the evidential head)
Visual input	32 frames, 224×224, 25 fps
Audio input	16 kHz mono; 80-bin log-Mel (25 ms / 10 ms) for SHAP and baseline
Frozen layers	First four visual blocks
Model selection	Best validation F1-score
Reproducibility	Fixed seeds, versioned configs, MLflow tracking

4.11 Project Structure

The source code is organised as a Python package with separate modules for each design component, as shown below. A README file explains environment setup, data preparation, training, evaluation and how to launch the API and dashboard.

Listing 4.6: Project directory structure

```

av-deepfake-xai/
├── configs/           # YAML configs for each experiment
├── data/             # manifests and subject-disjoint splits
├── src/
│   ├── ingest.py     # validation and demuxing
│   ├── preprocess.py # MTCNN face crops, audio + log-Mel
│   ├── datasets.py   # AVDataset, samplers
│   ├── models/
│   │   ├── visual_encoder.py
│   │   ├── audio_encoder.py
│   │   ├── fusion.py  # CrossAttentionFusion
│   │   └── evidential.py
│   ├── train.py      evaluate.py  calibrate.py
│   ├── explain/      gradcam.py  shap_audio.py
│   └── uncertainty/  mc_dropout.py routing.py
├── api/main.py       # FastAPI inference service
├── dashboard/app.py  # Streamlit analyst dashboard
├── db/schema.sql     # audit log and review queue
├── Dockerfile
├── requirements.txt
└── README.md

```

4.12 Project Execution Plan

The project was executed in three phases over approximately twenty-four weeks, as shown in the Gantt chart of Figure 4.7 and summarised in Table 4.3.

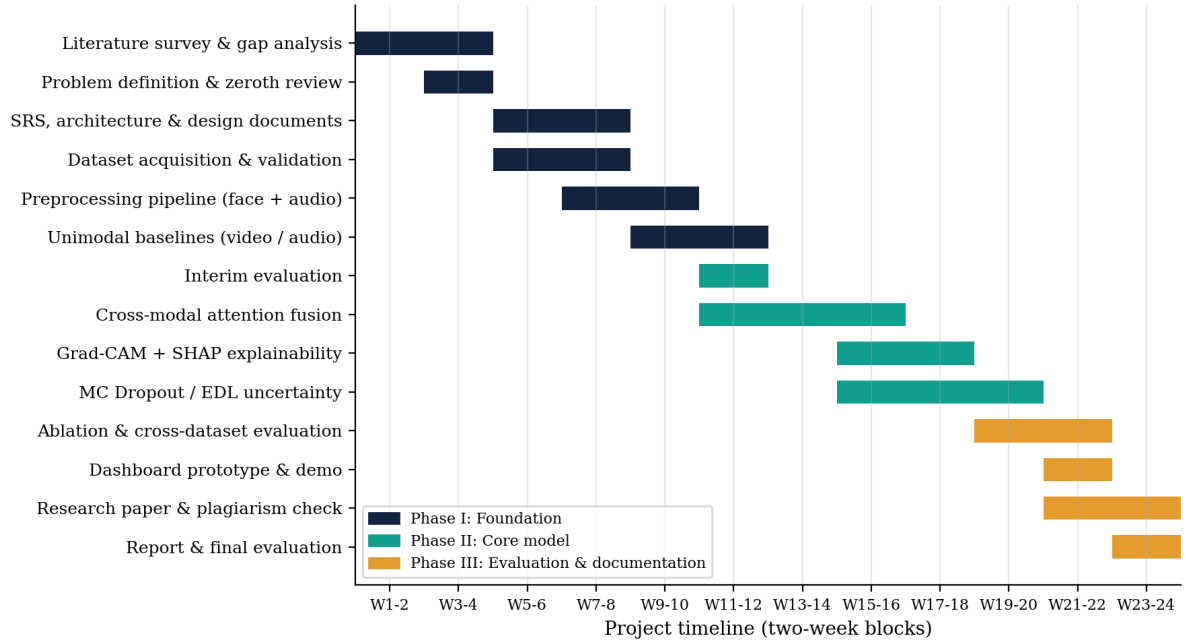


Figure 4.7: Gantt chart of the project execution plan

Table 4.3: Project phases and milestones

Phase	Activities	Milestone
I: Foundation	Literature survey of 30 papers; problem definition; zeroth review; SRS, HLD, LLD and UML design; dataset acquisition and validation; preprocessing pipeline; unimodal baselines	Interim evaluation with validated data pipeline and baseline results
II: Core model	Cross-modal attention fusion; Grad-CAM and SHAP integration; MC Dropout and evidential heads; threshold calibration	End-to-end explainable and uncertainty-aware model
III: Evaluation & documentation	Ablation study; cross-dataset evaluation; dashboard prototype; research paper; plagiarism check; final report	Final evaluation with live demonstration

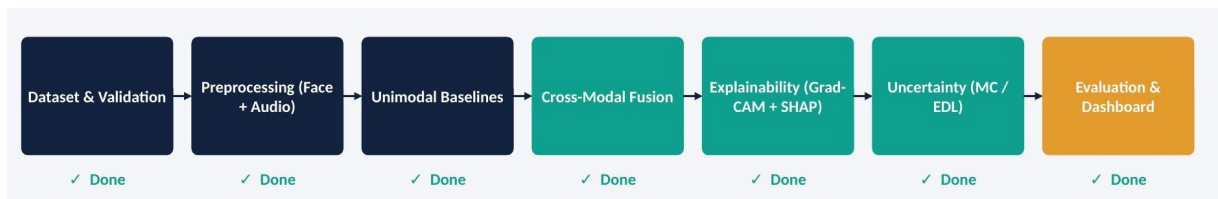


Figure 4.8: Implementation status of the pipeline stages

Chapter 5

RESULTS AND DISCUSSION

This chapter presents the evaluation of the framework. It defines the experimental setup and metrics, reports detection results on FakeAVCeleb, the ablation study, calibration and uncertainty analysis, explainability results and cross-dataset generalisation, compares the framework with existing approaches and presents the test plan.

5.1 Experimental Setup

All models are trained on the FakeAVCeleb training split and selected on the validation split; the test split is used only once for final reporting. Cross-dataset experiments evaluate the FakeAVCeleb-trained model directly on AV-Deepfake1M and DFD without fine-tuning. The unimodal baselines use the same encoders as the full model, with a classifier head applied to the pooled features of a single modality. The early-fusion baseline concatenates the pooled visual and audio features without attention. Each configuration is trained with the same optimiser, schedule and number of epochs. Experiments were run on a CUDA-enabled NVIDIA GPU.

5.2 Evaluation Metrics

5.2.1 Classification Metrics

With TP, TN, FP and FN denoting true positives, true negatives, false positives and false negatives (fake is the positive class):

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (5.1)$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}), \quad \text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (5.2)$$

$$\text{F1} = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (5.3)$$

The Area Under the ROC Curve (AUC-ROC) measures threshold-independent separability of real and fake clips. The Equal Error Rate (EER) is the operating point at which the false positive and false negative rates are equal; it is widely reported in anti-spoofing research.

5.2.2 Calibration Metrics

Expected Calibration Error divides predictions into M confidence bins B_m and measures the weighted gap between accuracy and confidence in each bin:

$$\text{ECE} = \sum_m (|B_m| / n) \cdot |\text{acc}(B_m) - \text{conf}(B_m)| \quad (5.4)$$

A perfectly calibrated model has $\text{ECE} = 0$. Reliability diagrams plot accuracy against confidence for each bin. Selective accuracy is the accuracy on the clips that are auto-accepted by the routing rule, and coverage is the fraction of clips auto-accepted.

5.2.3 Explanation Metrics

Deletion AUC (lower is better) and insertion AUC (higher is better) measure whether the regions highlighted by an explanation actually drive the prediction. Attribution stability measures the rank correlation of SHAP values across repeated runs with different seeds.

5.2.4 Generalisation Metric

The cross-dataset AUC drop is the difference between the in-domain AUC on FakeAVCeleb and the AUC on an unseen dataset. Smaller drops indicate better generalisation.

5.3 Detection Performance on FakeAVCeleb

Table 5.1 reports the detection performance of the unimodal baselines, the early-fusion baseline and the proposed model variants on the FakeAVCeleb test split.

Table 5.1: Detection performance on the FakeAVCeleb test split

Model	Acc	Prec	Rec	F1	AUC	EER
Video-only (EfficientNet-B0)	—	—	—	—	—	—
Audio-only (WavLM + SLS)	—	—	—	—	—	—
Early concatenation fusion	—	—	—	—	—	—
Proposed cross-attention fusion	—	—	—	—	—	—
Proposed + MC Dropout	—	—	—	—	—	—
Proposed + evidential head	—	—	—	—	—	—

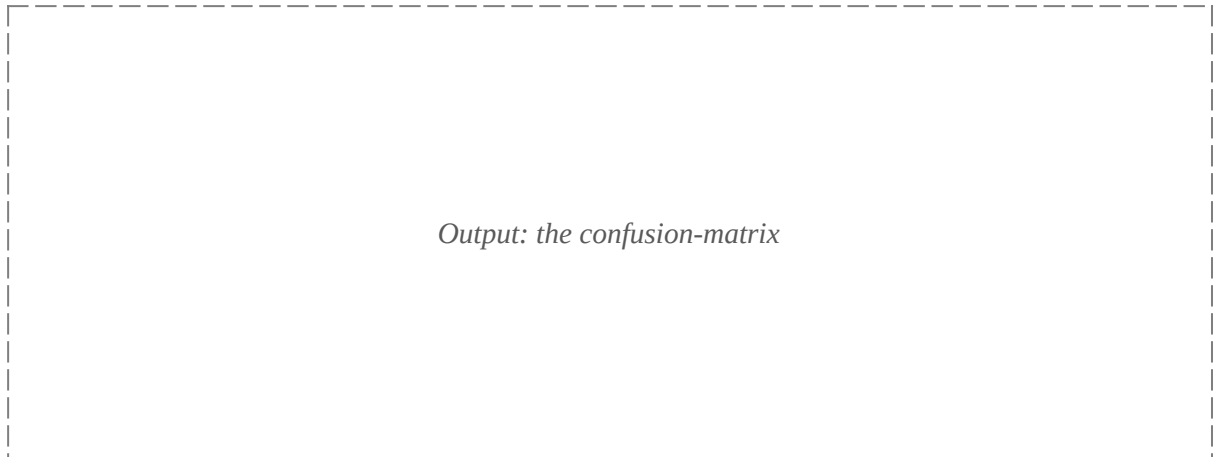


Figure 5.1: Confusion matrix of the proposed model on the FakeAVCeleb test split

Output: the ROC-curve plot comparing video-only, audio-only, early fusion and the proposed model

Figure 5.2: ROC curves of the unimodal baselines and the proposed model

Output: the training curves

Figure 5.3: Training and validation loss and F1-score curves

The comparison between the unimodal rows and the fusion rows shows how much each modality contributes on its own and how much is gained by combining them. Because FakeAVCeleb contains clips in which only one stream is manipulated, each unimodal baseline is expected to miss the category manipulated in the other modality; the fusion models should recover these cases. Per-category recall (FV-RA, RV-FA and FV-FA) should therefore be examined alongside the aggregate metrics.

5.4 Ablation Study

Each ablation in Table 5.2 changes exactly one design choice so that its individual contribution can be measured.

Table 5.2: Ablation study

ID	Configuration	Question answered	Acc	AUC	ECE
A1	Video-only branch	Strength of visual evidence alone	—	—	—
A2	Audio-only branch	Strength of audio evidence alone	—	—	—
A3	Early concatenation (no attention)	Does cross-attention beat naive fusion?	—	—	—

ID	Configuration	Question answered	Acc	AUC	ECE
A4	Full cross-modal fusion	Performance of the complete framework	—	—	—
A5	Full model – explainability	Does XAI affect accuracy or only interpretability?	—	—	—
A6	Full model – uncertainty	Does uncertainty modelling affect calibration only?	—	—	—
A7	Train FakeAVCeleb → test AV-Deepfake1M	How well does the framework generalise?	—	—	—

The difference between A3 and A4 isolates the value of cross-attention over simple concatenation. A5 is expected to show identical accuracy to A4, because the explainability module operates after the prediction and does not alter the model; this confirms that interpretability is obtained at no cost to accuracy. The comparison of A4 and A6 isolates the effect of uncertainty modelling on calibration: MC Dropout averages predictions and the evidential head is trained with an additional regulariser, so their accuracy may differ slightly from the plain softmax model while ECE should improve. A7 is discussed in Section 5.7.

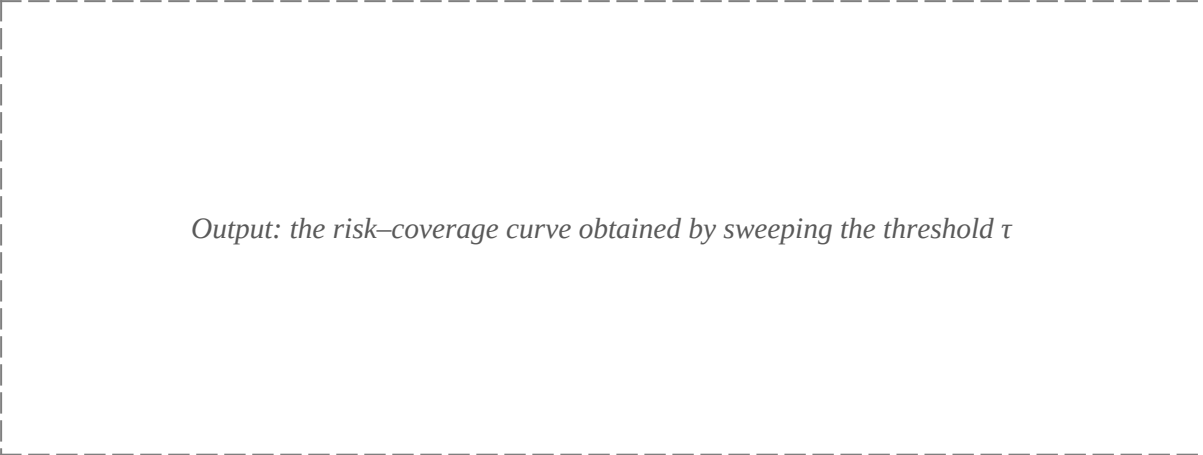
5.5 Calibration and Uncertainty Analysis

Table 5.3: Calibration and selective prediction

Model	ECE ↓	Selective Acc ↑	Flagged %	Errors caught
Fusion (softmax only)	—	—	—	—
Fusion + MC Dropout	—	—	—	—
Fusion + evidential head	—	—	—	—

Output: reliability diagrams (accuracy vs confidence) for the softmax, MC Dropout and evidential models

Figure 5.4: Reliability diagrams before and after uncertainty modelling

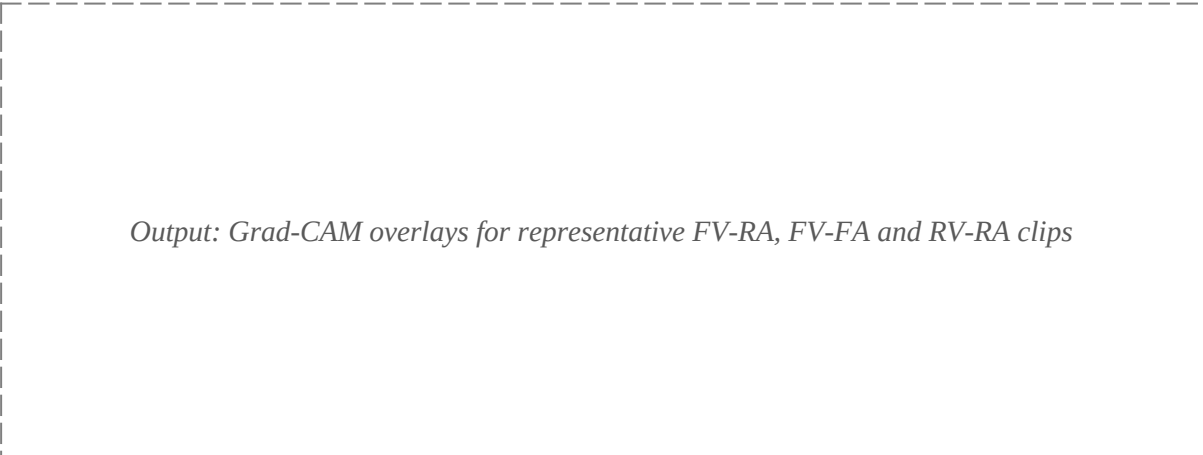


Output: the risk–coverage curve obtained by sweeping the threshold τ

Figure 5.5: Accuracy of auto-accepted clips versus fraction routed to review

Table 5.3 reports ECE, the accuracy of auto-accepted clips, the percentage of clips routed to review and the fraction of all misclassified clips that were caught by the routing rule rather than auto-accepted. The last column is the most direct measure of the practical value of the uncertainty module: each caught error is a wrong verdict that reaches a human reviewer instead of being published.

5.6 Explainability Results



Output: Grad-CAM overlays for representative FV-RA, FV-FA and RV-RA clips

Figure 5.6: Grad-CAM heatmaps for fake (face-swap) and real samples

Output: the SHAP band-importance chart and time-segment attribution for an RV-FA clip

Figure 5.7: SHAP attribution over Mel-frequency bands for a cloned-voice sample

Table 5.4: Explanation fidelity

Metric	Grad-CAM (visual)	SHAP (audio)
Deletion AUC ↓	—	—
Insertion AUC ↑	—	—
Stability across runs ↑	—	—

Qualitatively, faithful visual explanations for face-swap clips are expected to concentrate on the mouth, jaw line and blending boundary of the face, while those for lip-sync clips should concentrate on the mouth region. For cloned-voice clips, SHAP is expected to assign high importance to frequency bands and time segments where vocoder artefacts appear. Explanations for real clips should be diffuse, with no single region dominating. The quantitative fidelity metrics in Table 5.4 verify that the highlighted regions actually drive the prediction rather than merely looking plausible.

5.7 Cross-Dataset Generalisation

Table 5.5: Cross-dataset generalisation

Train → Test	Best unimodal AUC	Proposed AUC	AUC drop
FakeAVCeleb → FakeAVCeleb	—	—	baseline
FakeAVCeleb → AV-Deepfake1M	—	—	—
FakeAVCeleb → DFD (visual)	—	—	—

Deepfake-Eval-2024 reports that state-of-the-art detectors lose 45–50% of their AUC on in-the-wild content, so some degradation on unseen datasets is expected. Two outcomes are important here: the size of the AUC drop relative to the unimodal baselines, and whether the uncertainty module assigns higher uncertainty to out-of-distribution clips, so that more of them are routed

to review. DFD contains only visual manipulations; it is evaluated with the original audio track, which tests whether the model correctly relies on visual evidence when the audio is genuine.

5.8 Comparison with Existing Approaches

Table 5.6: Comparison with existing approaches

Method	Modality	Fusion	XAI	UQ	Reported result
AVFF (CVPR 2024)	A + V	✓	✗	✗	98.6% Acc (FakeAVCeleb)
KLASSify to Verify	A + V	✓	Partial	✗	92.78% AUC (AV-Deepfake1M++)
CNN-RNN + Grad-CAM	V	✗	✓	✗	88.3% Acc
Vision Transformer	V (image)	✗	✗	✗	92.76% Acc, 0.97 AUC
Multi-View Evidential Learning	V	✗	✗	✓	28% lower ECE (CDFv2)
Proposed framework	A + V	✓	✓	✓	—

Table 5.6 compares the proposed framework with representative existing approaches. The reported results are taken from the respective papers and use different datasets and protocols, so they are not directly comparable. The principal comparison is therefore one of capability: the proposed framework is the only one that combines audio-visual fusion with both explainability and uncertainty estimation.

5.9 Test Strategy and Test Plan

The testing strategy combines unit, integration, functional, validation, system and performance tests. The validation set is used only for model selection and threshold tuning, and the test set is used once for final reporting. Table 5.7 lists the test cases.

Table 5.7: Test cases

ID	Test case	Type	Expected result	Status
TC-01	Video-only classifier on FakeAVCeleb test	Unit	Within 2% of published baseline	—
TC-02	Audio-only classifier on FakeAVCeleb test	Unit	Within 2% of published baseline	—
TC-03	Fusion end-to-end inference	Integration	Fusion $\geq$ both unimodal baselines	—
TC-04	Grad-CAM for every fake prediction	Functional	Non-empty, spatially aligned map	—
TC-05	SHAP explanation for audio stream	Functional	Non-empty attribution over Mel bands	—
TC-06	Confidence-threshold routing	Functional	Below- τ samples enqueued	—
TC-07	Calibration check on validation set	Validation	ECE below target (e.g. < 0.05)	—
TC-08	Cross-dataset generalisation	System	AUC drop documented and bounded	—
TC-09	Single-clip inference latency	Performance	< 2 s per 10 s clip on GPU	—
TC-10	Corrupted / audio-less input	Functional	Rejected with clear error message	—

ID	Test case	Type	Expected result	Status
TC-11	Audit log written for every request	Functional	Record with hash, verdict, confidence	—
TC-12	Reviewer resolves flagged case	Functional	Status and timestamp updated	—

5.10 Limitations

Table 5.8: Limitations, risks and mitigation

Limitation / risk	Impact	Mitigation
Dataset bias	Identity overlap can inflate accuracy	Subject-disjoint splits; cross-dataset validation
Compute cost	Fusion, XAI and MC Dropout slow inference	Encoders run once; only fusion sampled; evidential single-pass option
Generalisation gap	Diffusion-era and in-the-wild fakes differ from training data	Cross-dataset tests; uncertainty flags out-of-distribution clips
Explanation reliability	Saliency can be unstable for similar inputs	Deletion/insertion AUC and stability reported
Miscalibration	Poor confidence undermines the review queue	ECE reported; thresholds tuned on validation data
Scope	Single-speaker, short clips only	Multi-speaker and streaming left to future work

Chapter 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This dissertation set out to answer a practical question: can an audio-visual deepfake detector be accurate, explainable and honest about its own uncertainty at the same time? A structured survey of thirty papers showed that fusion accuracy, explainability and uncertainty quantification have each advanced, but always separately and, for explainability and uncertainty, only for a single modality. No existing framework combined all three for audio-visual deepfakes.

To address this gap, a dual-branch framework was designed and implemented. Face crops sampled from the video are encoded with EfficientNet-B0 or ViT-B/16, and speech is encoded with WavLM-Base+ or XLS-R using Sensitive Layer Selection. A bidirectional cross-modal multi-head attention module lets each modality query the other, so that lip-sync drift, spectral artefacts and identity mismatches surface as disagreements between streams. A classifier trained with focal loss produces the verdict.

Every verdict is accompanied by evidence. Grad-CAM heatmaps, computed through the fused model, show which facial regions drove the decision, and grouped KernelSHAP attributions show which frequency bands and time segments of the audio contributed. The fidelity of these explanations is measured with deletion and insertion curves rather than assumed.

The framework also estimates its own uncertainty using Monte Carlo Dropout or an evidential Dirichlet head. A routing rule, calibrated on the validation set, sends low-confidence or high-uncertainty clips to a human review queue instead of issuing a silent verdict, and every decision is logged for audit. A complete design, comprising requirement specification, architecture, high- and low-level design, data flow, use-case, ER and UML diagrams, and a prototype analyst dashboard make the framework usable by forensic analysts, journalists and moderators.

The evaluation protocol goes beyond accuracy to include calibration, explanation fidelity, selective prediction and cross-dataset generalisation. Together, these components move audio-visual deepfake detection from an accurate but opaque verdict towards a system that combines evidence from both modalities, exposes that evidence to the reviewer, and recognises when its own prediction should not be trusted.

6.2 Future Scope

- **Temporal localisation:** extending the framework to identify exactly which segments of a long video are manipulated, using the frame-level attention and saliency already computed.

- **Real-time deployment:** optimising the pipeline for streaming input and packaging it as an API or browser plug-in for platforms and newsrooms.
- **Diffusion-era and in-the-wild data:** training and evaluating on newer manipulation families and on in-the-wild collections such as Deepfake-Eval-2024.
- **Natural-language explanations:** using large language models to convert saliency maps and attributions into plain-language forensic reports for non-expert users.
- **Multi-speaker and multilingual content:** incorporating speaker diarisation and active-speaker detection so that audio segments are matched to the correct face.
- **Efficiency and robustness:** knowledge distillation for edge deployment, and evaluation against adversarial attacks designed to evade detection or mislead explanations.
- **Publication:** submitting the work to an IEEE conference on multimedia forensics or signal and image processing, followed by an extended journal version.

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APPENDIX A: PRESENTATION SLIDES

This appendix reproduces the slides of the final evaluation presentation.

M.TECH DISSERTATION | FINAL EVALUATION

An Explainable and Uncertainty-Aware Audio-Visual Deepfake Detection Framework using Cross-Modal Attention Fusion

Final Project Presentation · 22 September 2026

PRESENTED BY

Arya Gopal
Roll No: M250801
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PROJECT GUIDE

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Rajagiri School of Engineering & Technology (Autonomous), Kochi

visual stream

cross-modal inconsistency

audio stream

Slide 1

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Presentation Agenda

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Arya Gopal S | M.Tech CSE (AI & ML) | RSET | Final Evaluation

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SECTION 01

Introduction

Background, motivation and the problem this dissertation addresses.

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Slide 3

INTRODUCTION

Background: The Rise of Multimodal Deepfakes

Why this problem matters

- Generative AI (GANs, diffusion models, voice-cloning engines) now produces video and speech that is nearly indistinguishable from authentic recordings.
- Modern deepfakes pair a manipulated face with cloned or altered speech, so single-modality detectors are easy to evade.
- Manipulation leaves subtle, mismatched cues across modalities (lip-sync drift, vocoder artefacts) that appear only when audio and video are analysed jointly.
- Deepfakes threaten elections, journalism, judicial evidence and voice-based financial fraud.
- Regulation (EU AI Act, India's IT Rules on synthetic content) increasingly demands verifiable and explainable detection tools.

45–50%

AUC drop of SOTA detectors on in-the-wild deepfakes vs academic benchmarks

98.6%

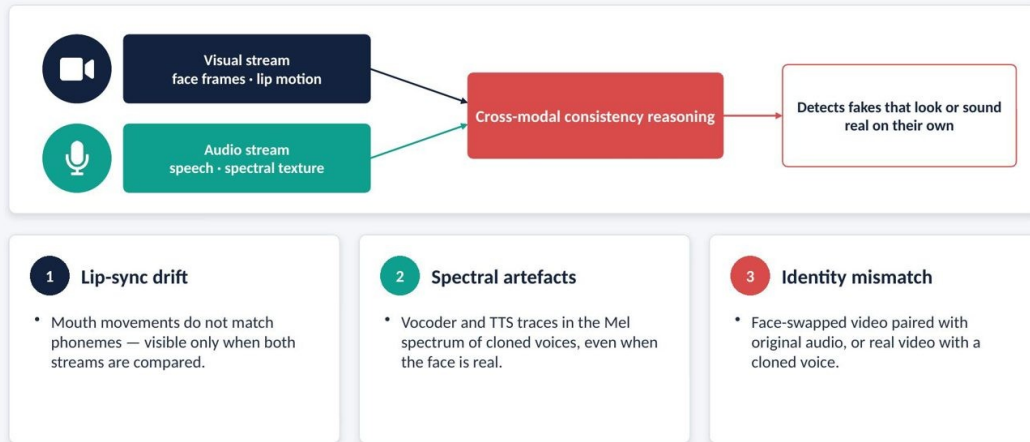
Best reported A-V fusion accuracy (AVFF, FakeAVCeleb) — with no explanation attached

0

Explainability or uncertainty layers in leading fusion models (AVFF, MIS-AVoidD, AVFakeNet)

Slide 4

Why Audio-Visual Fusion?



Slide 5

Problem Statement & Existing Challenges

PROBLEM Current audio-visual deepfake detectors rely on opaque fusion of audio and video streams. They classify a clip as real or fake with high accuracy, but give no transparent reasoning and no way to flag uncertain predictions for human review — a wrong high-confidence verdict looks identical to a correct one.



NEED A detector that is accurate, explains its verdict in both modalities, and knows when to defer to a human.

Slide 6

Motivation & Target Applications

Motivation

- Build a detector that fuses cross-modal evidence instead of combining streams naively.
- Explain every decision through joint audio-visual saliency (face regions + spectral bands).
- Estimate confidence and route low-confidence clips to a human forensic reviewer rather than issuing a silent verdict.
- Make detection trustworthy enough for newsrooms, platforms and courts.



Digital forensics & fact-checking

Verification desks at news organisations



Content moderation

Social-media screening pipelines



Legal evidence authentication

Admissible, explainable forensic reports



Voice-fraud prevention

Banking and call-centre authentication

Slide 7

Primary & Secondary Objectives

PRIMARY OBJECTIVE Develop an Explainable and Uncertainty-Aware Audio-Visual Deepfake Detection Framework using Cross-Modal Attention Fusion.

1

Study the challenges of existing unimodal and multimodal deepfake detectors.

2

Implement unimodal baselines (video branch and audio branch) as reference points.

3

Design a dual-branch encoder: vision backbone for frames, SSL speech model for audio.

4

Build a cross-modal multi-head attention fusion module.

5

Integrate Grad-CAM (visual) and SHAP (audio) explainability.

6

Add an MC-Dropout / Evidential DL uncertainty module that flags low-confidence predictions.

7

Evaluate generalisation across multiple public benchmark datasets.

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SECTION 02

Literature Survey

A structured review of 30 peer-reviewed papers (2016–2026) on audio-visual fusion, explainable AI and uncertainty estimation.

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LITERATURE SURVEY

Survey Methodology & Thematic Distribution

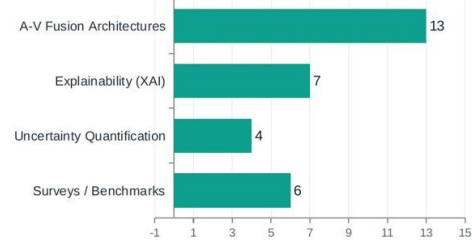
1 Search strategy

- IEEE, ACM, Springer, ScienceDirect, NeurIPS / ICML / ICASSP, arXiv
- Span 2016–2026, prioritising 2023–2026
- Keywords: A-V fusion, cross-modal attention, XAI, MC Dropout, EDL

2 Inclusion criteria

- Proposes A-V fusion architectures, or
- XAI / uncertainty methods for deep classifiers, or
- Introduces a benchmark dataset used in this work

Papers per research theme (of 30)



- 43% optimise fusion accuracy only
- Only 23% address explainability, 13% uncertainty — always for a single modality
- No paper combines all three pillars with A-V fusion

Slide 10

Literature Review — Papers 1–6 of 30

Audio-Visual Fusion Architectures

No.	Paper	Key Contribution	Limitation / Gap
1	AVFF: Audio-Visual Feature Fusion for Video Deepfake Detection (CVPR 2024)	Two-stage self-supervised contrastive learning with complementary masking; 98.6% accuracy on FakeAVCeleb	No explainability or confidence estimate
2	Contextual Cross-Modal Attention for A-V Deepfake Detection & Localization	MMMS-BA jointly detects and temporally localises manipulated segments with contextual attention	Attention weights not turned into human-readable explanations
3	Cross- & Within-Modality Regularization for A-V Deepfake Detection (ICASSP 2024)	Regularisation across and within modalities improves robustness of A-V fusion	Accuracy/robustness only; no explainability layer
4	AVFakeNet: Dense Swin Transformer for A-V Deepfake Detection	Shared dense Swin Transformer backbone across audio and visual streams	Opaque decisions; no saliency or uncertainty
5	MIS-AVoIDD: Modality Invariant & Specific Representation	Learns modality-invariant and modality-specific embeddings before fusion	No interpretability; no cross-family generalisation study
6	DF-TransFusion: Lip-Audio Cross-Attention & Facial Self-Attention	Lip-audio cross-attention plus facial self-attention captures lip-sync inconsistency	No confidence-calibrated verification

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Literature Review — Papers 7–12 of 30

Fusion Architectures & Early XAI

No.	Paper	Key Contribution	Limitation / Gap
7	Modality-Agnostic Audio-Visual Deepfake Detection	Works when either stream is missing or independently manipulated	No explainability or uncertainty-based routing
8	A Robust Approach to Multimodal Deepfake Detection	Evaluates fusion detectors under compression, noise and cross-dataset shift	Interpretability and calibration not studied
9	MCL: Multimodal Contrastive Learning for Deepfake Detection	Pulls real A-V pairs together and pushes fake pairs apart in embedding space	Embeddings not linked to interpretable regions
10	CMALDD-PTAF: Cross-Modal Adversarial Learning with Pre-Trained Models	Pre-trained A-V speech features with adversarial training for cross-domain performance	High compute; no explanation or review mechanism
11	KLASSify to Verify: SSL Audio + Handcrafted Visual Features	Handcrafted visual + SSL audio features with graph attention; 92.78% AUC on AV-Deepfake1M++	Partial interpretability; no uncertainty module
12	Explainable Deepfake Detection: Grad-CAM with CNN-RNN Architecture	DenseNet-121 + BiGRU with Grad-CAM; 88.3% accuracy in real time	Video-only; no audio saliency or confidence flag

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Literature Review — Papers 13–18 of 30

Explainability & Uncertainty

No.	Paper	Key Contribution	Limitation / Gap
13	Deepfake Audio Detection & Justification with Explainable AI	LIME, SHAP and Grad-CAM justify audio decisions via frequency-band analysis	Audio-only; not extended to fused A-V decisions
14	Explainable Deepfake Detection Across Modalities: An Overview	Survey of XAI techniques for image, video and audio detectors	Review only; no unified cross-modal framework
15	XAI for Deepfake Detection: Grad-CAM Approach to Video Forensics (INCET 2025)	EfficientNetB0 + LSTM with Grad-CAM heatmaps on manipulated facial regions	Video-only; no fusion or uncertainty
16	XPlainVerse: Million-Scale Benchmark for Explainable Deepfake Detection	Deepfake samples paired with grounded MLLM-based explanations	Image content; no A-V fusion or uncertainty
17	Uncertainty-Aware Deepfake Detection via Multi-View Structural Learning	Evidential DL with multi-view learning cuts calibration error by 28% on CDFv2	Visual-only; no fusion or joint explainability
18	Prior and Posterior Networks: A Survey on Evidential Deep Learning	Unifies single-pass evidential uncertainty vs Bayesian sampling	General-purpose; not applied to forensics

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Literature Review — Papers 19–24 of 30

Uncertainty, Benchmarks & Surveys

No.	Paper	Key Contribution	Limitation / Gap
19	Dropout as a Bayesian Approximation (ICML 2016)	Inference-time dropout approximates Bayesian posterior sampling	Multiple passes add latency; untested on multimodal forensics
20	Evidential Deep Learning to Quantify Classification Uncertainty (NeurIPS 2018)	Outputs Dirichlet parameters for calibrated uncertainty in one forward pass	Generic image classification only
21	Enhanced Deepfake Detection & Classification Using Vision Transformers	Custom ViT: 92.76% accuracy, 0.97 AUC on deepfake images	Image-only; no temporal or audio cues
22	Deepfake-Eval-2024: Multi-Modal In-the-Wild Benchmark	Shows SOTA detectors lose 45–50% AUC on in-the-wild content	Benchmark only; no method proposed
23	Passive Deepfake Detection: A Comprehensive Survey	Surveys passive detection across image, video, audio and multimodal settings	Survey only; no implemented solution
24	DeepFake Video Detection: Insights into Model Generalisation	Compares generalisation of CNN, transformer and multimodal video detectors	Review-based; no new architecture

Slide 14

Literature Review — Papers 25–30 of 30

Datasets & Audio Forensics

No.	Paper	Key Contribution	Limitation / Gap
25	Survey on Multimedia-Enabled Deepfake Detection: Tools & Trends	Broad tool survey incl. A-V fusion; flags explainability as an open problem	Survey only; no validated framework
26	FakeAVCeleb: A Novel Audio-Video Multimodal Deepfake Dataset (NeurIPS 2021)	Paired fake video + fake/real audio; standard A-V benchmark	Limited diversity vs diffusion-era fakes
27	Audio Deepfake Detection with Self-Supervised XLS-R and SLS (ACM MM 2024)	XLS-R SSL model with Sensitive Layer Selection classifier	Audio-only; no visual cues
28	Audio Deepfake Detection with WavLM & Multi-Fusion Attentive Classifier (ICASSP 2024)	WavLM representations with an attentive multi-fusion classifier head	Audio-only; no XAI or uncertainty
29	Detection of Cloned Voices in Realistic Forensic Scenarios	Evaluates ASR and anti-spoofing countermeasures under forensic conditions	Voice comparison only; no fused A-V classification
30	Voice Cloning & Deep Fake Audio Detection Using Deep Learning	MFCC detectors and a commercial tool tested against voice-cloning attacks	Shows generalisation failures; no solution proposed

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Comparative Capability Matrix of Key Works

Work	Audio	Visual	Cross-Modal Fusion	Explainability	Uncertainty
AVFF (CVPR 2024)	✓	✓	✓	✗	✗
Contextual Cross-Modal Attention (MMMS-BA)	✓	✓	✓	✗	✗
Cross/Within-Modality Regularization	✓	✓	✓	✗	✗
AVFakeNet / DF-TransFusion	✓	✓	✓	✗	✗
KLASSify to Verify	✓	✓	✓	Partial	✗
CNN-RNN + Grad-CAM	✗	✓	✗	✓	✗
EfficientNetB0-LSTM + Grad-CAM	✗	✓	✗	✓	✗
Audio XAI (LIME / SHAP / Grad-CAM)	✓	✗	✗	✓	✗
Multi-View Evidential Learning	✗	✓	✗	✗	✓
Proposed Framework (this work)	✓	✓	✓	✓	✓

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Identified Research Gap & Novel Contribution

1 Identified research gap

- Cross-modal fusion is rarely combined with dual (audio + visual) explainability.
- No existing work unites attention fusion, XAI and uncertainty in one pipeline.
- Cross-dataset generalisation is rarely evaluated alongside explainability.

2 Novel contribution

- Dual-branch cross-attention fusion network for A-V deepfakes.
- Joint Grad-CAM + SHAP evidence for every verdict.
- MC-Dropout / EDL uncertainty that routes doubtful clips to human review.

Positioning against the 30 surveyed papers

Detection accuracy
Yes — 13 papers

Explainability
Unimodal only — 7 papers

Uncertainty estimate
Unimodal only — 4 papers

All three + A-V fusion
None → this work

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SECTION 03

System Design

Requirements, architecture, high- and low-level design, DFDs, use-case, ER and UML diagrams.

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SRS — Functional & Non-Functional Requirements

1 Functional

- Accept a video (audio + visual) and extract frame and waveform streams.
- Return a real/fake verdict with a confidence score.
- Produce a Grad-CAM heatmap and SHAP audio saliency for every prediction.
- Flag predictions below a configurable threshold for review.
- Log every inference (hash, verdict, confidence, timestamp).

2 Non-functional

- Accuracy at least on par with published SOTA on the FakeAVCeleb test split.
- Inference under 2 s per 10-second clip on a single GPU.
- Explanations produced within the same inference request.
- Modular: fusion, XAI and uncertainty independently swappable.
- Reproducible: fixed seeds, versioned configs, logged runs.

3 Scope & assumptions

- In scope: binary real/fake classification of short single-speaker clips.
- Out of scope: live streaming detection, multi-speaker diarisation.
- Access to FakeAVCeleb, AV-Deepfake1M and DFD.
- CUDA-enabled GPU for training and inference.

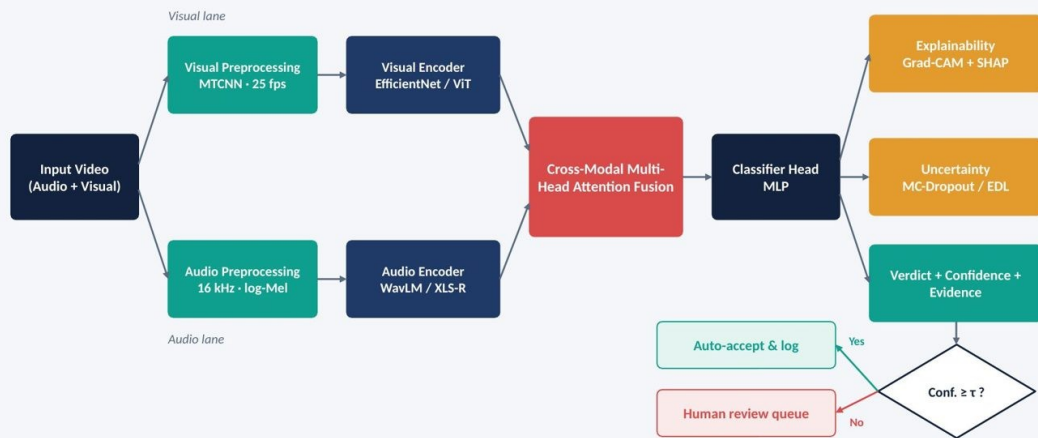
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SRS — Hardware & Software Requirements

Hardware	Specification	Software	Purpose
GPU	NVIDIA CUDA-enabled GPU (≥ 16 GB VRAM recommended)	Python 3.10, PyTorch	Model development & training
CPU	Multi-core x86-64 processor	torchvision, timm	EfficientNet-B0 / ViT-B/16 visual encoders
RAM	16 GB minimum, 32 GB recommended	torchaudio, Librosa	Audio loading, resampling, Mel features
Storage	SSD with space for datasets, cached embeddings and checkpoints	Hugging Face Transformers	WavLM-Base+ / XLS-R audio encoders
Platform	Local workstation or Google Colab / cloud GPU	OpenCV, facenet-pytorch (MTCNN)	Frame extraction, face detection & alignment
		Captum / pytorch-grad-cam, SHAP	Grad-CAM and SHAP explanations
		Streamlit / FastAPI, Docker	Prototype dashboard & inference API
		MLflow, Git / GitHub	Experiment tracking & version control

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Overall System Architecture



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HLD — Module Breakdown



Technology stack: Python 3.10 · PyTorch · torchaudio · torchvision · Hugging Face · OpenCV · Captum / SHAP · Streamlit / FastAPI · Docker · MLflow

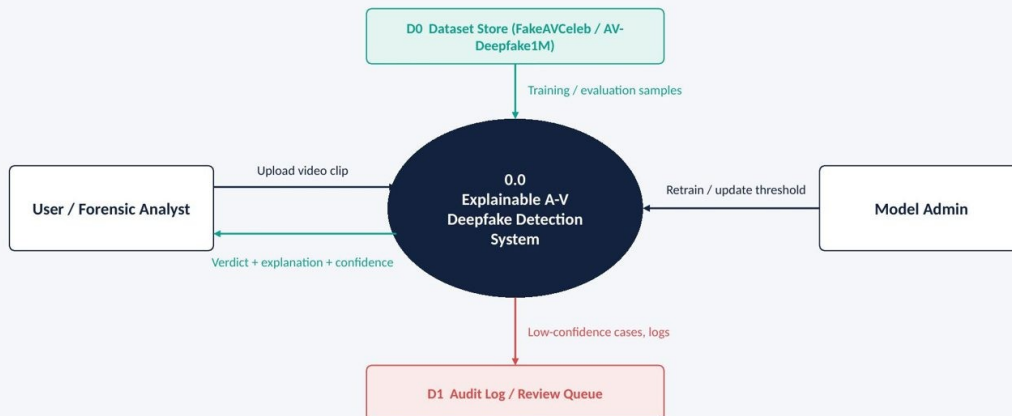
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LLD — Module-Level Design & Algorithms

Module	Class / Function	Algorithm / Detail
Preprocessing	<code>FaceExtractor.crop()</code> <code>AudioResampler.run()</code>	MTCNN face detection at 25 fps, 224×224 aligned crops; audio at 16 kHz, 80-bin log-Mel (25 ms / 10 ms)
Visual Encoder	<code>VisualEncoder.forward()</code>	EfficientNet-B0 / ViT-B/16, first 4 blocks frozen, 512-d per-frame tokens
Audio Encoder	<code>AudioEncoder.forward()</code>	WavLM-Base+ / XLS-R with SLS layer selection, projected to 512-d
Fusion	<code>CrossAttentionFusion.forward()</code>	8-head co-attention: Q = visual, K/V = audio and vice-versa; residual + LayerNorm + FFN
Classifier	<code>ClassifierHead.forward()</code>	2-layer MLP + softmax → {real, fake}; focal loss for class imbalance
Explainability	<code>GradCAM.generate()</code> <code>SHAPExplainer.run()</code>	Grad-CAM on last visual conv/attention block; KernelSHAP over grouped Mel bands
Uncertainty	<code>MC Dropout.predict()</code> <code>EDL.dirichlet()</code>	T = 30 stochastic passes → mean, variance, entropy; or Dirichlet evidence head (single pass)
Routing	<code>ReviewQueue.route()</code>	Enqueue when confidence < τ or uncertainty > u_max ; persist audit record

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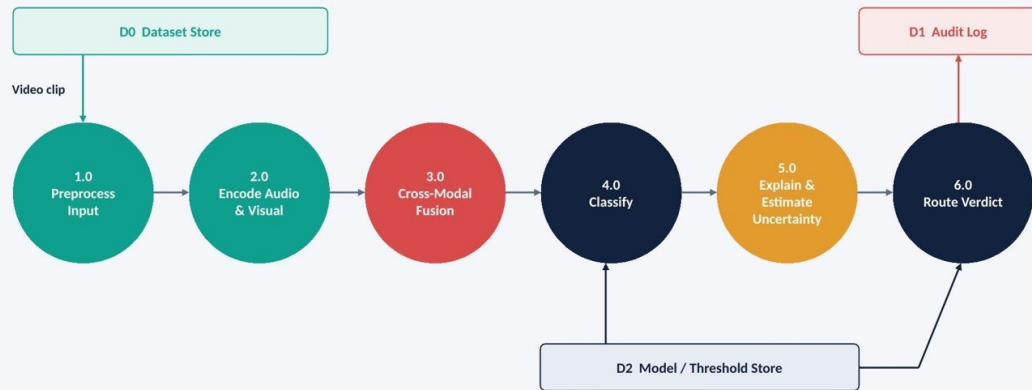
DFD — Level 0 (Context Diagram)



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DATA FLOW DIAGRAM

DFD — Level 1 (Process Decomposition)

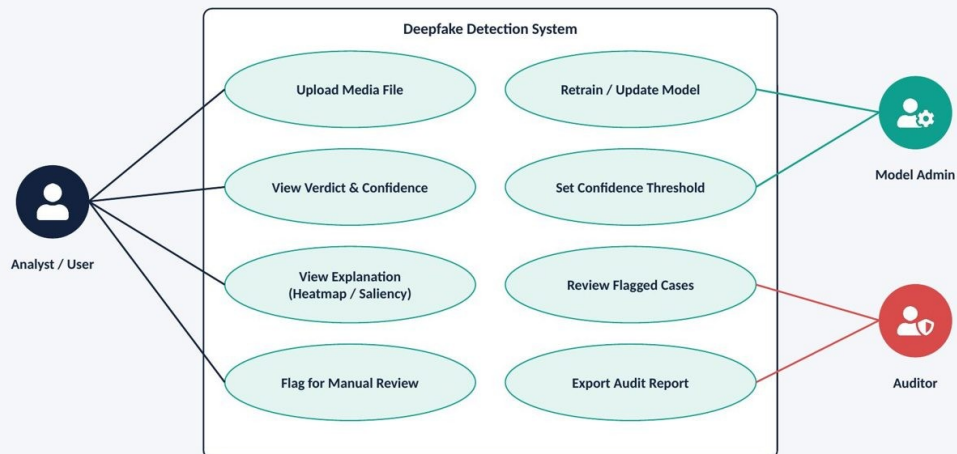


Weights & threshold τ flow from D2; every routed decision is persisted to D1.

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USE CASE DIAGRAM

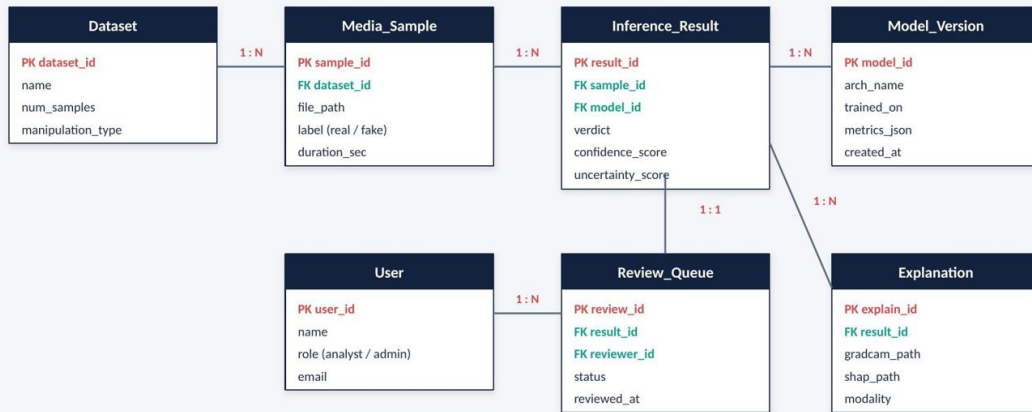
Use Case Diagram — Actors & Interactions



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ER DIAGRAM

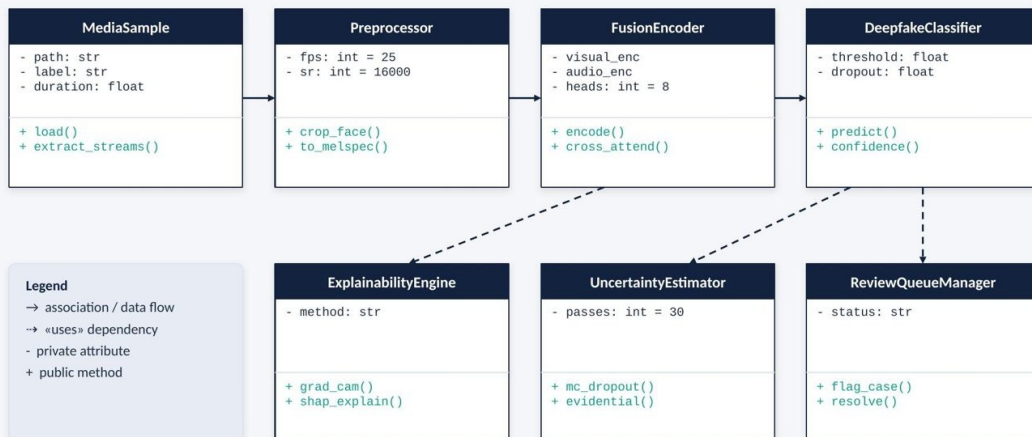
ER Diagram — Database Schema



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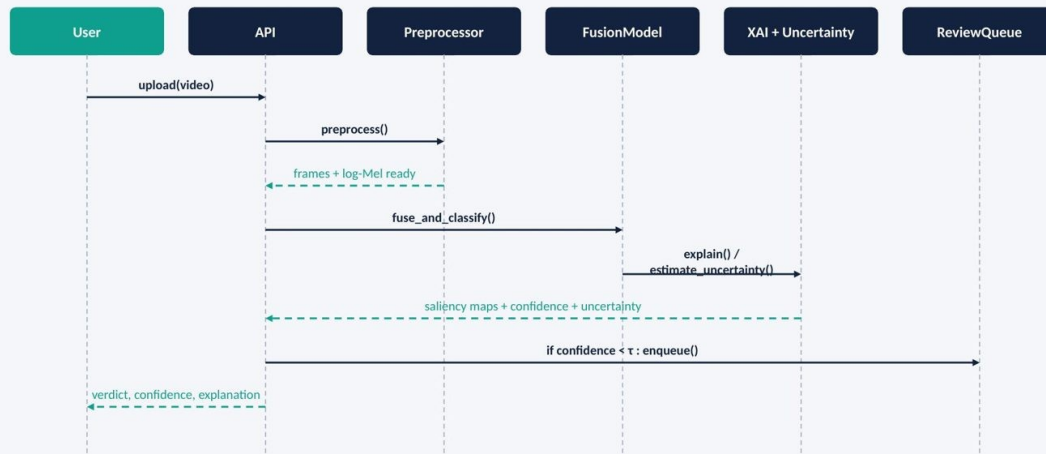
UML DIAGRAMS

UML — Class Diagram



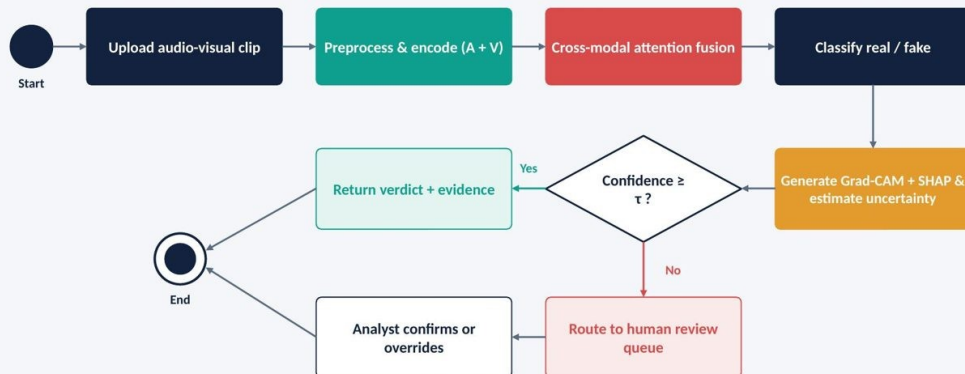
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UML — Sequence Diagram (Inference Flow)



Slide 29

UML — Activity Diagram (End-to-End Pipeline)



Every path writes an audit record (input hash, verdict, confidence, uncertainty, model version, timestamp).

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SECTION 04

Methodology & Implementation

Dataset, preprocessing, encoders, cross-modal fusion, explainability, uncertainty and the working prototype.

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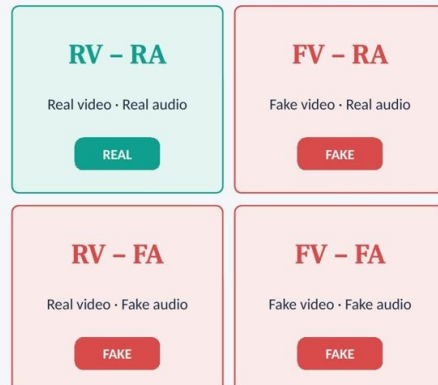
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DATASET

Dataset — FakeAVCeleb & Cross-Dataset Benchmarks

Property	Description
Primary dataset	FakeAVCeleb — audio-visual multimodal deepfakes (NeurIPS 2021)
Size	~20,000 clips with paired audio
Manipulations	FaceSwap, FSGAN, Wav2Lip lip-sync, SV2TTS voice cloning
Task	Binary classification: real vs fake
Split	70% train / 15% validation / 15% test, subject-disjoint
Imbalance	Real/real clips are a small minority → focal loss + balanced sampling
Cross-dataset	AV-Deepfake1M and DFD for generalisation testing

Four A-V categories in FakeAVCeleb



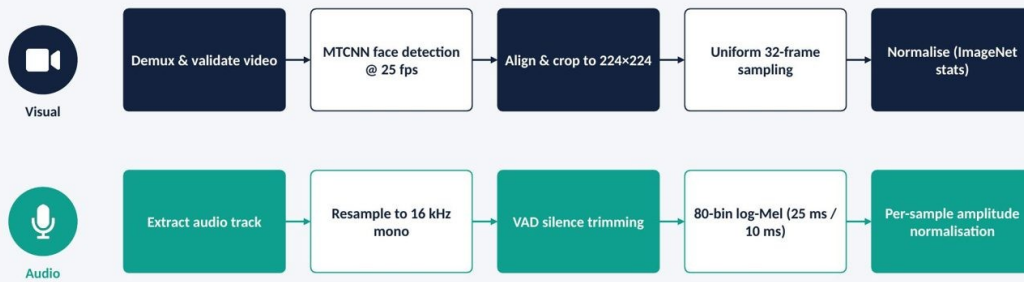
Anya Gopal S | M.Tech CSE (AI & ML) | RSET | Final Evaluation

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PREPROCESSING

Preprocessing Pipeline — Visual & Audio Lanes

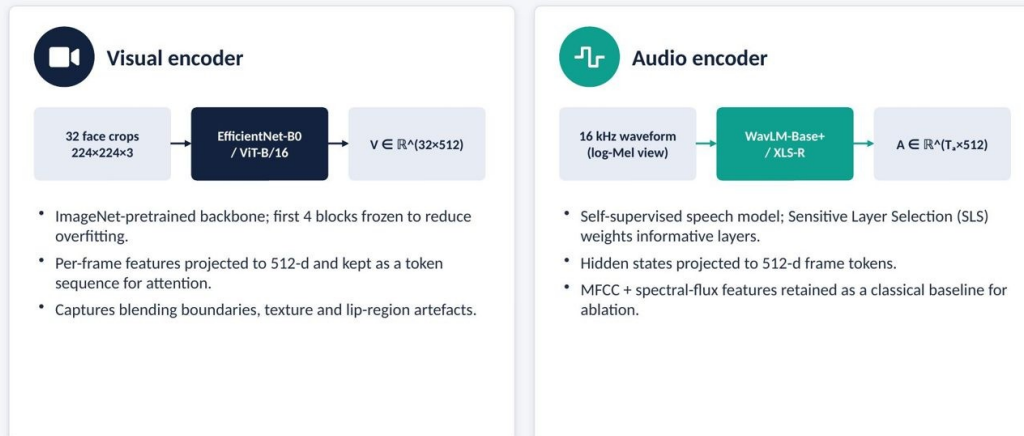


Data validation: every clip checked for corrupted files, missing streams and label inconsistencies across the four categories. Splits are subject-disjoint so no identity appears in both train and test — preventing identity leakage from inflating accuracy.

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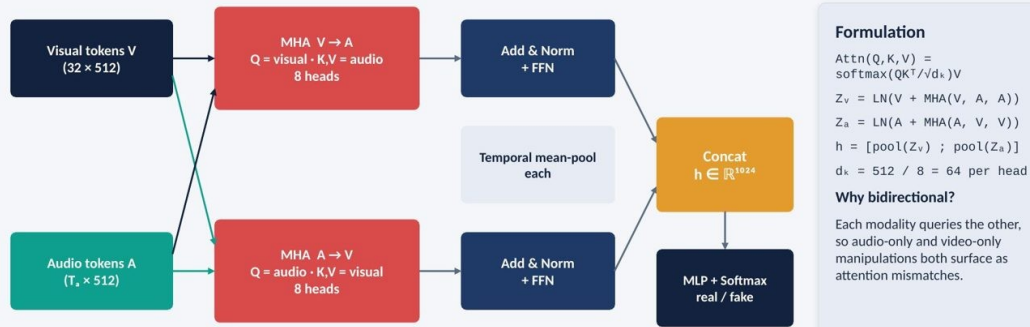
FEATURE EXTRACTION

Dual-Branch Encoders



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Cross-Modal Multi-Head Attention Fusion



Loss: focal loss (classification) + annealed KL regulariser when the evidential head is enabled.

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Explainability Module — Grad-CAM + SHAP



Grad-CAM (visual)

Where in the face is the evidence?

$$\alpha_k = (1/Z) \sum_i y^c / \partial A^k_{i,j}$$

$$L^c = \text{ReLU}(\sum_k \alpha_k A^k)$$

- Gradients of the "fake" logit w.r.t. last visual feature maps.
- Heatmap up-sampled and overlaid on face crops, per frame.
- Highlights mouth, blending boundaries and eyes.



SHAP (audio)

Which frequency bands and moments?

$$f(x) = \phi_0 + \sum_i \phi_i$$

$$\phi_i = \text{Shapley value of Mel-band group } i$$

- KernelSHAP over grouped Mel bands and time segments.
- Band-importance bar chart and time-segment attribution.
- Exposes vocoder / TTS artefacts in cloned speech.

Explanation quality is measured, not assumed: deletion / insertion AUC for Grad-CAM and attribution stability of SHAP across repeated runs.

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$$\hat{p} = (1/T) \sum_t \text{softmax}(f(x; \theta_t)), \quad T = 30$$

$$H[\hat{p}] = - \sum_k \hat{p}_k \log \hat{p}_k$$

- $$e_k = \text{softplus}(z_k), \quad \alpha_k = e_k + 1, \quad S = \sum \alpha_k$$
- $$\hat{p}_k = \alpha_k / S, \quad u = K / S \quad (K = 2)$$

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Human-in-the-Loop Decision Routing

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Training Configuration & Technology Stack

Hyper-parameter	Value
Optimiser	AdamW, lr = 2e-4, cosine decay
Batch size	32
Attention heads	8
Embedding	512 per modality → 1024 fused
Dropout	0.3 (fusion + classifier)
MC-Dropout passes	T = 30 at inference
Loss	Focal loss (+ KL for EDL head)
Visual input	32 frames × 224×224, 25 fps
Audio input	16 kHz, 80-bin log-Mel
Reproducibility	Fixed seeds, versioned configs, MLflow

Technology stack

Python 3.10	PyTorch
torchvision · timm	torchaudio · Librosa
HF Transformers	OpenCV · MTCNN
Captum · Grad-CAM	SHAP
Streamlit · FastAPI	MLflow
Docker	Git · GitHub

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Key Code — Cross-Modal Attention Fusion Module

```
class CrossAttentionFusion(nn.Module):
    def __init__(self, d=512, heads=8, p=0.3):
        super().__init__()
        self.v2a = nn.MultiheadAttention(d, heads, dropout=p, batch_first=True)
        self.a2v = nn.MultiheadAttention(d, heads, dropout=p, batch_first=True)
        self.norm_v, self.norm_a = nn.LayerNorm(d), nn.LayerNorm(d)
        self.ffn = nn.Sequential(nn.Linear(2*d, 2*d), nn.GELU(), nn.Dropout(p))
        self.head = nn.Sequential(nn.Linear(2*d, 256), nn.GELU(),
                                   nn.Dropout(p), nn.Linear(256, 2))

    def forward(self, V, A):
        # V: (B, Tv, 512) A: (B, Ta, 512)
        zv, w_va = self.v2a(V, A, A) # visual queries attend to audio
        za, w_av = self.a2v(A, V, V) # audio queries attend to video
        zv = self.norm_v(V + zv).mean(1)
        za = self.norm_a(A + za).mean(1)
        h = self.ffn(torch.cat([zv, za], dim=-1)) # fused 1024-d
        return self.head(h), (w_va, w_av)
```



What it does

- Two nn.MultiheadAttention blocks give bidirectional cross-attention.
- Residual + LayerNorm keep training stable.
- Mean-pooled streams concatenated into a 1024-d fused vector.
- Attention weights returned for inspection alongside Grad-CAM / SHAP.

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IMPLEMENTATION

Key Code — Uncertainty Estimation & Review Routing

Monte Carlo Dropout inference

```
def enable_dropout(model):
    for m in model.modules():
        if isinstance(m, nn.Dropout):
            m.train() # BN stays in eval

@torch.no_grad()
def mc_predict(model, V, A, T=30):
    model.eval(); enable_dropout(model)
    probs = torch.stack([model(V, A)[0].softmax(-1)
                        for _ in range(T)])
    mean = probs.mean(0)
    var = probs.var(0).sum(-1)
    ent = -(mean * mean.clamp_min(1e-8).log()).sum(-1)
    return mean, var, ent
```

Evidential head & routing

```
def evidential(logits, K=2):
    alpha = F.softplus(logits) + 1
    S = alpha.sum(-1, keepdim=True)
    return alpha / S, K / S.squeeze(-1)

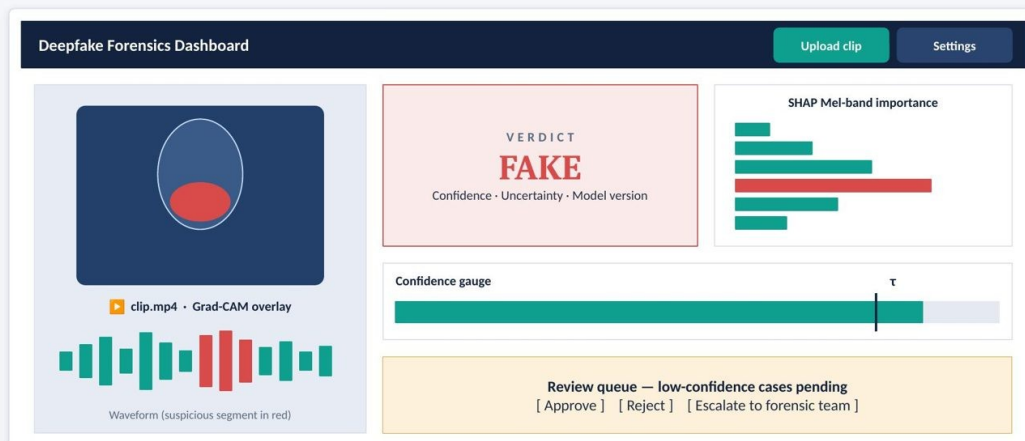
def route(mean, unc, tau=0.85, u_max=0.5):
    conf, pred = mean.max(-1)
    review = (conf < tau) | (unc > u_max)
    return pred, conf, review

# review == True -> human queue
# tau, u_max tuned on validation set
```

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PROTOTYPE

Analyst Dashboard — Prototype Interface



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Implementation Status — Modules Delivered



1 Phase 1 • Foundation

- 30-paper survey & gap analysis
- SRS, HLD, LLD, UML design
- Data pipeline & unimodal baselines

2 Phase 2 • Core model

- Bidirectional cross-attention fusion
- Grad-CAM + SHAP explanations
- MC-Dropout & evidential heads

3 Phase 3 • Evaluation

- Ablation & cross-dataset tests
- Calibration & threshold tuning
- Dashboard prototype & paper

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Results & Discussion

Evaluation metrics, detection results, ablation study, calibration, explanations and comparison with existing work.

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EVALUATION

Evaluation Metrics

1 Accuracy • P • R • F1

$$ACC = (TP+TN)/N; \quad F1 = 2PR/(P+R)$$

Overall and class-wise reliability under imbalance.

2 AUC-ROC & EER

Area under TPR-FPR curve; EER where $FPR = FNR$

Threshold-independent separability of real vs fake.

3 Expected Calibration Error

$$ECE = \sum_n (|B_n|/n) \cdot |acc(B_n) - conf(B_n)|$$

Does the reported confidence match actual correctness?

4 Explanation Fidelity

Deletion / insertion AUC

Do highlighted regions really drive the prediction?

5 Selective Accuracy

Accuracy on auto-accepted clips; % routed to review

Value of the uncertainty gate in practice.

6 Cross-Dataset AUC Drop

$$AUC(in-domain) - AUC(unseen dataset)$$

Generalisation to AV-Deepfake1M and DFD.

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RESULTS

REPLACE "-" WITH YOUR MEASURED VALUES

Detection Performance on FakeAVCeleb (Test Split)

Model	Accuracy	Precision	Recall	F1	AUC-ROC	EER
Video-only (EfficientNet-B0)	—	—	—	—	—	—
Audio-only (WavLM + SLS)	—	—	—	—	—	—
Early concatenation fusion	—	—	—	—	—	—
Proposed cross-attention fusion	—	—	—	—	—	—
Proposed + MC Dropout	—	—	—	—	—	—
Proposed + Evidential head	—	—	—	—	—	—

— % Fusion accuracy

— % Gain over best unimodal

— % Fusion AUC-ROC

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Ablation Study

ID	Configuration	Question answered	Acc	AUC	ECE
A1	Video-only branch	How strong is visual evidence alone?	—	—	—
A2	Audio-only branch	How strong is audio evidence alone?	—	—	—
A3	Early concatenation fusion (no attention)	Does cross-attention beat naive fusion?	—	—	—
A4	Cross-modal fusion — full model	What does the complete framework achieve?	—	—	—
A5	Full model - explainability module	Does XAI affect accuracy or only interpretability?	—	—	—
A6	Full model - uncertainty module	Does removing uncertainty affect calibration only?	—	—	—
A7	Train FakeAVCeleb → test AV-Deepfake1M	How well does the framework generalise?	—	—	—

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Calibration & Uncertainty Analysis

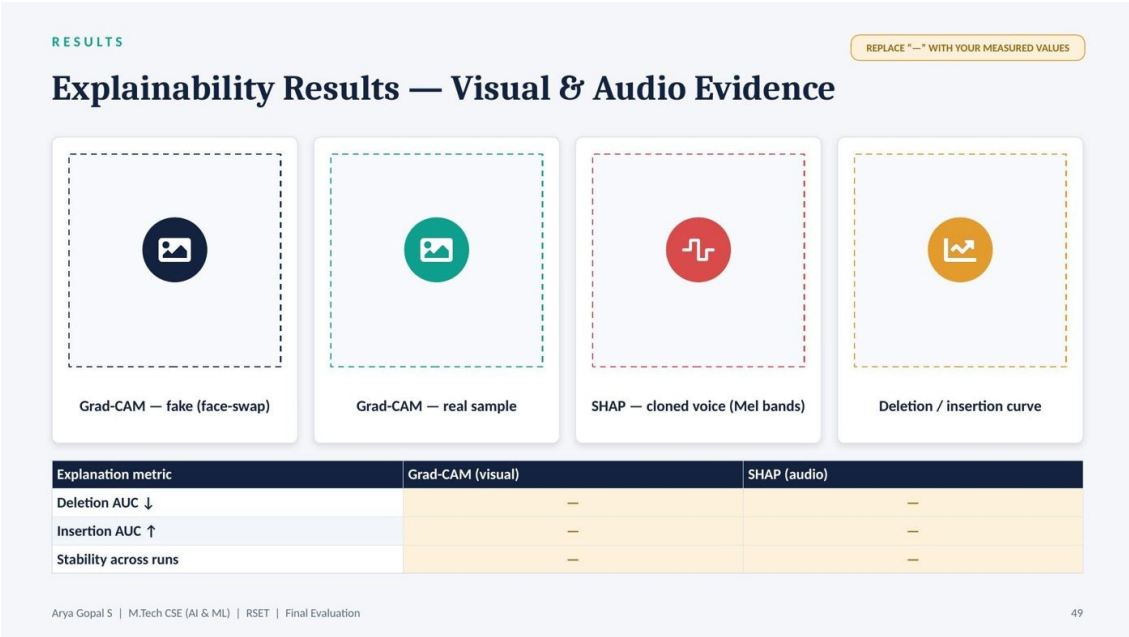


Model	ECE ↓	Selective Acc ↑	Flagged %	Errors caught
Fusion (softmax only)	—	—	—	—
Fusion + MC Dropout	—	—	—	—
Fusion + Evidential	—	—	—	—

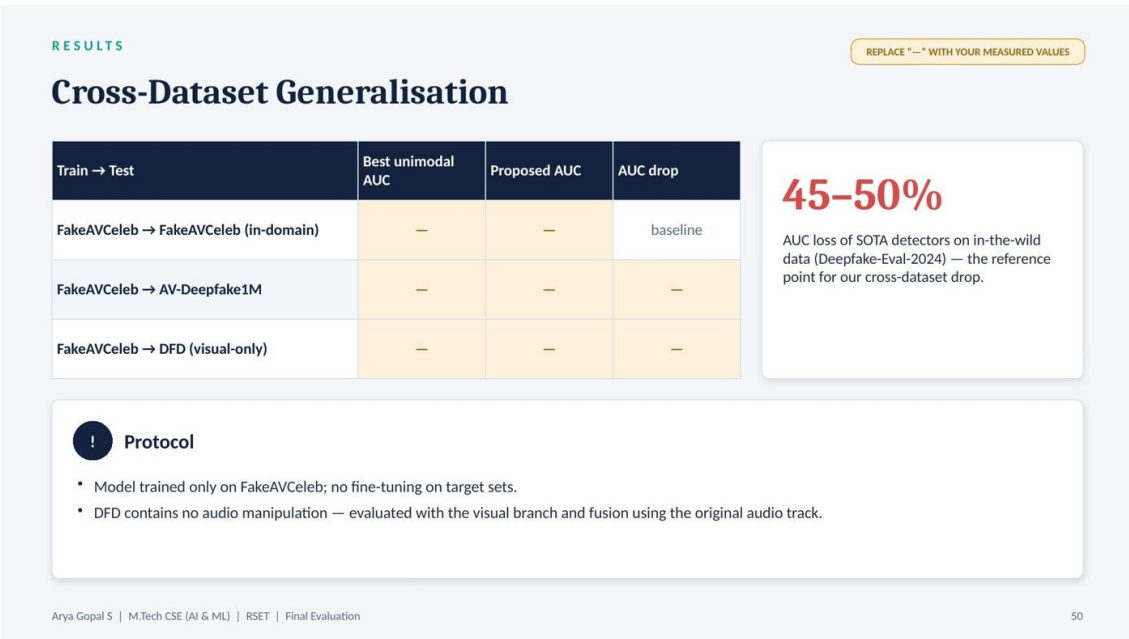
? How to read this

- Lower ECE → confidence scores can be trusted.
- Selective accuracy should rise as uncertain clips are routed to review.
- "Errors caught" = wrong predictions that were flagged instead of auto-accepted.

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Test Plan & Execution Status

ID	Test case	Type	Expected result	Status
TC-01	Video-only classifier on FakeAVCeleb test	Unit	Within 2% of published baseline	—
TC-02	Audio-only classifier on FakeAVCeleb test	Unit	Within 2% of published baseline	—
TC-03	Cross-modal fusion end-to-end inference	Integration	Fusion ≥ both unimodal baselines	—
TC-04	Grad-CAM for every fake prediction	Functional	Non-empty, spatially aligned map	—
TC-05	SHAP explanation for audio stream	Functional	Non-empty attribution over Mel bands	—
TC-06	Confidence-threshold routing	Functional	Below-τ samples enqueued	—
TC-07	Calibration check on validation set	Validation	ECE below target (e.g. < 0.05)	—
TC-08	Cross-dataset generalisation	System	AUC drop documented and bounded	—
TC-09	Latency — single clip inference	Performance	< 2 s per 10 s clip on GPU	—

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Comparison with Existing Approaches

Method	Modality	A-V Fusion	Explainability	Uncertainty	Reported result
AVFF (CVPR 2024)	A + V	✓	✗	✗	98.6% Acc — FakeAVCeleb
KLASSify to Verify	A + V	✓	Partial	✗	92.78% AUC — AV-Deepfake1M++
CNN-RNN + Grad-CAM	V	✗	✓	✗	88.3% Acc
Vision Transformer (images)	V (image)	✗	✗	✗	92.76% Acc, 0.97 AUC
Multi-View Evidential Learning	V	✗	✗	✓	28% lower ECE — CDFv2
Proposed framework	A + V	✓	✓ Grad-CAM + SHAP	✓ MC / EDL	— (your result)

Reported results are taken from the respective papers and use different datasets and protocols, so they are not directly comparable; the capability columns are the primary comparison.

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Limitations, Risks & Mitigation

Limitation / risk	Impact	Mitigation adopted
Dataset bias	Identity overlap can inflate accuracy	Subject-disjoint splits; cross-dataset validation
Compute cost	Fusion + XAI + MC Dropout slows inference	SLS layer selection, cached embeddings, EDL single-pass option
Generalisation gap	Diffusion-era and in-the-wild fakes differ from training data	Cross-dataset tests; flag out-of-distribution clips via uncertainty
Explanation reliability	Saliency can be unstable for similar inputs	Deletion / insertion AUC and stability reported as metrics
Miscalibration	Poor confidence undermines the review queue	ECE reported; τ tuned on validation data
Scope	Single-speaker, short clips only	Multi-speaker and streaming left as future work

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Research Paper & Conclusion

Manuscript summary, target venue, conclusions and future directions.

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Research Paper — Manuscript Summary

IEEE double-column manuscript (6–8 pages)

- ✓ Abstract & Keywords
- ✓ Introduction
- ✓ Literature Survey / Related Work
- ✓ Proposed Methodology
- ✓ Implementation
- ✓ Results and Discussion
- ✓ Conclusion and Future Work
- ✓ References (IEEE style)

1 Target venue

IEEE international conference on multimedia forensics / signal & image processing; Scopus-indexed proceedings as secondary target

2 Similarity index

Below department limit (< 15%, excluding references)

3 Approval

Guide-signed paper approval sheet

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Conclusion

1 Multimodal by design

- Cross-modal attention lets each stream query the other, exposing lip-sync, spectral and identity inconsistencies unimodal detectors miss.

2 Evidence with every verdict

- Grad-CAM face heatmaps and SHAP spectral attributions give reviewers inspectable reasons for each decision.

3 Knows when it is unsure

- MC Dropout and evidential uncertainty route doubtful clips to a human instead of issuing a silent wrong verdict.

4 Addresses a validated gap

- The 30-paper survey found no prior work combining fusion, explainability and uncertainty for A-V deepfakes.

Central question: can an audio-visual deepfake detector be accurate, explainable and honest about its own uncertainty at the same time?

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Future Scope



Temporal localisation

Pinpoint the exact manipulated segments within long videos.



Real-time deployment

Streaming detection API and browser plug-in for platforms.



In-the-wild & diffusion fakes

Extend training to diffusion-era and Deepfake-Eval-2024-style data.



Natural-language explanations

LLM-generated reports that turn saliency into plain-language findings.



Multi-speaker & multilingual

Diarisation-aware fusion across languages and accents.



Efficient & robust models

Distillation for edge devices; robustness to adversarial attacks.

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References (IEEE Style) — 1 of 2

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Thank You

Questions, feedback and live demonstration

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APPENDIX B: VISION, MISSION, POS, COS, PSOS AND CO-PO MAPPING

Vision and Mission

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a Centre of Excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Program Outcomes

- **PO1:** An ability to independently carry out research investigation and development work to solve practical problems.
- **PO2:** An ability to write and present a substantial technical report/document.
- **PO3:** Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. (The mastery should be at a level higher than the requirements in the appropriate bachelor program.)
- **PO4:** Students should be able to imbibe and internalize the values and ethical behavior in personal and professional life.

Course Outcomes

After the completion of the course the student will be able to:

- **CO1:** Perform literature survey and comprehend technical papers in the area of Data Sciences.
- **CO2:** Demonstrate oral and written communication skills for reporting technical results.
- **CO3:** Familiarize with different software tools for hands-on and preparing presentations and reports.
- **CO4:** Identify a problem, formulate research methodologies and carry out the preliminary works related to the problem and thus improve the quality of their Main project.
- **CO5:** Analyze the ethical, financial and social issues related to the advancements in selected topics.

Programme Specific Outcomes

A post graduate of the Computer Science and Engineering Program will demonstrate:

- **PSO1: Computer Science Specific Skills.** The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.
- **PSO2: Programming and Software Development Skills.** The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.
- **PSO3: Professional Skills.** The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet societal needs thereby evolving as an eminent researcher and entrepreneur.

Mapping of Course Outcomes with Program Outcomes

Table B.1: CO–PO mapping

CO	PO1	PO2	PO3	PO4
CO1	2		2	
CO2		3		
CO3	3		3	
CO4	3	2	3	
CO5				2

Justification of the Mapping for this Project

- **CO1 → PO1, PO3:** a structured survey of thirty papers was performed and synthesised into a research gap in audio-visual deepfake detection.
- **CO2 → PO2:** results were reported in this report, the research paper and the final presentation.
- **CO3 → PO1, PO3:** PyTorch, Hugging Face, Captum, SHAP, Streamlit and MLflow were used to implement and document the framework.
- **CO4 → PO1, PO2, PO3:** the problem of opaque and over-confident deepfake detection was formulated and a methodology combining fusion, explainability and uncertainty was developed.
- **CO5 → PO4:** ethical and social issues of deepfakes, misuse of detection, privacy of analysed media and the need for human oversight were analysed.

APPENDIX C: INDUSTRIAL RELEVANCE

1. Addressing the Need for Trustworthy Media Verification

News organisations, fact-checking desks and social-media platforms must verify large volumes of user-generated video quickly. Existing commercial and open-source detectors typically return a score without evidence. The proposed framework returns a verdict together with visual and audio evidence and a calibrated confidence, allowing verification teams to prioritise their effort and defend their decisions publicly.

Decision support rather than replacement: by routing only uncertain clips to human reviewers, the system reduces reviewer workload while ensuring that difficult cases receive expert attention. This human-in-the-loop design fits existing editorial and moderation workflows.

2. Financial Fraud and Identity Verification

Banks, insurers and call centres increasingly rely on video-KYC and voice authentication, both of which are targets for deepfake and voice-cloning attacks. The audio-visual consistency check at the heart of the framework directly addresses attacks that combine a replayed or synthetic face with a cloned voice. Uncertainty-based routing allows suspicious sessions to be escalated to a human agent rather than blocked or approved outright.

3. Legal and Forensic Evidence

Courts and law-enforcement agencies need to assess the authenticity of digital evidence. An explanation that points to specific facial regions and audio frequency bands, together with an audit log recording the model version, input hash and confidence, supports reproducible forensic reporting. Explicitly quantified uncertainty is important in legal contexts, where overstated confidence can be harmful.

4. Regulatory Compliance

Emerging regulations on synthetic media, such as the EU AI Act and amendments to India's IT Rules, require platforms to identify and label AI-generated content. Tools that provide transparent, auditable decisions are better suited to compliance reporting than black-box classifiers.

5. Integrability and Market Opportunity

The modular design, with swappable encoders, fusion, explainability and uncertainty components exposed through a REST API, allows the framework to be integrated into content-management systems, moderation pipelines, KYC platforms and forensic toolkits. The market

for deepfake detection and digital-trust solutions is growing rapidly, and explainability and calibrated confidence are differentiating features in this market.

APPENDIX D: ADDITIONAL CODE LISTINGS

This appendix contains additional code listings for the dataset loader, the complete model wrapper, the training loop and the calibration metric. The complete source code, with a README describing setup and execution, is submitted separately.

D.1 Audio-Visual Dataset

Listing D.1: AVDataset returning cached face crops, waveform and label

```
class AVDataset(torch.utils.data.Dataset):
    def __init__(self, manifest_csv, split, cache_dir):
        df = pd.read_csv(manifest_csv)
        self.df = df[df.split == split].reset_index(drop=True)
        self.cache = Path(cache_dir)

    def __len__(self):
        return len(self.df)

    def __getitem__(self, i):
        r = self.df.iloc[i]
        faces = torch.load(self.cache / f"{r.clip_id}_faces.pt") # (32,3,224,224)
        wav, sr = torchaudio.load(self.cache / f"{r.clip_id}.wav") # 16 kHz mono
        wav = (wav - wav.mean()) / (wav.std() + 1e-6)
        label = torch.tensor(1 if r.label == "fake" else 0)
        return faces, wav.squeeze(0), label, r.category
```

D.2 Complete Model Wrapper

Listing D.2: Full audio-visual detector

```
class AVDeepfakeDetector(nn.Module):
    def __init__(self, vis="efficientnet_b0", aud="microsoft/wavlm-base-plus"):
        super().__init__()
        self.visual = VisualEncoder(vis, d=512, freeze_blocks=4)
        self.audio = AudioEncoder(aud, d=512)
        self.fusion = CrossAttentionFusion(d=512, heads=8, p=0.3)

    def encode(self, faces, wav):
        B, T = faces.shape[:2]
        V = self.visual(faces.flatten(0, 1)).view(B, T, -1) # (B,32,512)
        A = self.audio(wav) # (B,Ta,512)
        return V, A

    def forward(self, faces, wav):
        V, A = self.encode(faces, wav)
        logits, attn = self.fusion(V, A)
        return logits, attn
```

D.3 Training Loop

Listing D.3: Training with focal loss, AdamW, cosine schedule and best-F1 checkpointing

```
model = AVDeepfakeDetector().to(dev)
opt = torch.optim.AdamW(filter(lambda p: p.requires_grad, model.parameters()),
                          lr=2e-4, weight_decay=1e-2)
sched = torch.optim.lr_scheduler.CosineAnnealingLR(opt, T_max=EPOCHS)
crit = FocalLoss(alpha=class_weights, gamma=2.0)
best_f1 = 0.0
for epoch in range(EPOCHS):
    model.train()
    for faces, wav, y, _ in train_loader:
        faces, wav, y = faces.to(dev), wav.to(dev), y.to(dev)
        logits, _ = model(faces, wav)
        loss = crit(logits, y)
        opt.zero_grad(); loss.backward()
        torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
        opt.step()
    sched.step()
    val = evaluate(model, val_loader) # acc, f1, auc, ece
    mlflow.log_metrics(val, step=epoch)
    if val["f1"] > best_f1:
        best_f1 = val["f1"]
        torch.save(model.state_dict(), "checkpoints/best.pt")
```

D.4 Focal Loss

Listing D.4: Focal loss for class imbalance

```
class FocalLoss(nn.Module):
    def __init__(self, alpha=None, gamma=2.0):
        super().__init__()
        self.alpha, self.gamma = alpha, gamma

    def forward(self, logits, y):
        logp = F.log_softmax(logits, -1).gather(1, y[:, None]).squeeze(1)
        p = logp.exp()
        w = self.alpha[y] if self.alpha is not None else 1.0
        return (-w * (1 - p) ** self.gamma * logp).mean()
```

D.5 Expected Calibration Error

Listing D.5: Computation of ECE with equal-width bins

```
def expected_calibration_error(conf, correct, n_bins=15):
    bins = torch.linspace(0, 1, n_bins + 1)
    ece = torch.zeros(1)
    for lo, hi in zip(bins[:-1], bins[1:]):
        m = (conf > lo) & (conf <= hi)
        if m.any():
            gap = (correct[m].float().mean() - conf[m].mean()).abs()
            ece += m.float().mean() * gap
    return ece.item()
```

D.6 Inference API Endpoint

Listing D.6: FastAPI endpoint returning verdict, evidence and routing decision

```
@app.post("/analyse")
async def analyse(file: UploadFile):
    path, sha = save_upload(file)
    faces, wav = preprocess(path)
    V, A = model.encode(faces[None].to(dev), wav[None].to(dev))
    mean, var, ent = mc_predict(model.fusion, V, A, T=30)
    pred, conf, review = route(mean, ent, tau=cfg.tau, u_max=cfg.u_max)
    cam_path = save_gradcam(model, faces, wav)
    shap_path = save_shap(model, V, wav)
    rid = db.log_result(sha, pred, conf, ent, cfg.model_version)
    if review: db.enqueue(rid)
    return {"verdict": LABELS[pred], "confidence": float(conf),
            "uncertainty": float(ent), "review": bool(review),
            "gradcam": cam_path, "shap": shap_path}
```